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Machine learning-driven forecasting of composite wholesale price indices for agricultural commodities in the Vegetable Basket Project

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Abstract
Projecting composite wholesale agricultural commodity price indices within the Vegetable Basket Project equips policymakers and market participants with empirical tools for optimizing production scheduling, procurement decision-making, and risk management by forecasting volatility and directional shifts. Such precise index estimation enhances supply chain efficiency and reduces financial risk through informed hedging strategies. Here, we introduce a novel forecasting architecture employing Gaussian process regression (*GPR*) optimized within a Bayesian inference framework, enabling the model to adjust to latent market forces and sudden structural transitions. Integrating time-varying dynamics allows the algorithm to more faithfully reproduce nonstationary price trajectories. The empirical component analyzes daily observations spanning September 27, 2005–April 10, 2025, encompassing eras of regulatory change, sectoral transformation, and macroeconomic upheaval. In an out-of-sample validation from February 17, 2022 to April 10, 2025, our GPR model achieves a relative root mean square error (*RRMSE*) of 0.2040 percent, a *RMSE* of 0.4841, a *MAE* of 0.3430, and a correlation coefficient (*CC*) of 0.99929. To the best of our awareness, this research constitutes the inaugural implementation of a Bayesian-tuned *GPR* methodology for composite wholesale agricultural price forecasting. By contributing to machine learning-based predictive theory, this flexible framework can be adapted to similarly complex time-series tasks in other commodity domains.

Keywords Composite wholesale agricultural commodity price index · Vegetable Basket Project · Time-series forecasting · Gaussian regression process · Bayesian optimization · Cross validation

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1 Introduction

Projecting composite wholesale agricultural price indices of the Vegetable Basket Project in China serves a pivotal role across economic planning, market functioning, welfare policies, and regulatory design by equipping diverse actors–from smallholder cultivators and agribusiness firms to governmental bodies and financial institutions–with the ability to forecast price trajectories, mitigate exposure, and allocate inputs with heightened accuracy (Chen et al. 2012; Zhang and Na 2018; Sediyono et al. 2025; Panda et al. 2024). By synthesizing price movements of staples such as grains, oilseeds, fruits, and vegetables into a unified metric, researchers can filter out idiosyncratic volatility present in individual commodity quotations and unveil deeper structural patterns in supply and demand that might otherwise be concealed by episodic disturbances, thus delivering a more reliable foundation for strategic decision-making than isolated price indicators (Mitra and Paul 2017; Zhong et al. 2020; Xu and Zhang 2024; Sun et al. 2023; Xu and Zhang 2022; Kurumatani 2020). At the producer level, precise anticipations of the composite index of the Vegetable Basket Project enable optimization of crop portfolios, planting calendars, and resource deployment–ranging from fertilizer and irrigation to labor–by synchronizing field operations with likely market conditions, which not only boosts yield efficiency but also curtails the financial ramifications of price downturns (Ding and Zhang 2020; Shahwan and Odening 2007; Xu 2018; Tran et al. 2023; Xu 2018; Wihartiko et al. 2021; Xu and Zhang 2022). Such proactive adjustments help minimize surplus accumulation–with attendant storage expenses and spoilage risks–and prevent shortages that elevate procurement costs downstream in the value chain (Manogna et al. 2025; Jin and Xu 2025q; Zhong et al. 2021; Jin and Xu 2025w; Islam et al. 2023; Jin and Xu 2025aa; Mohanty et al. 2023; Xu 2019; Rl and Mishra 2021). For intermediaries and logistics operators, dynamic index forecasts support adaptive inventory control and routing schedules that lower carrying costs and streamline reverse logistics, facilitate just-in-time distribution, and reinforce coordination among warehouses, hubs, and retail points–collectively strengthening supply chain resilience and reducing end-user prices (Liu et al. 2023; Xu and Hsu 2022; Drachal 2019; Yuan and Ling 2020; Xu 2017; Mahto et al. 2021). Agribusinesses evaluating capital deployment in processing plants, cooling systems, and value-added facilities rely on projected price levels and volatilities to inform discounted cash flow models, thereby guiding investment timing and scale decisions that foster private participation in modernizing China's agricultural infrastructure (An et al. 2023; Bayona-Oré et al. 2021; Bundak et al. 2025; Utomo et al. 2022; Hegde et al. 2020; Zhong et al. 2023; Xu 2019; Zhao 2021; Xu and Thurman 2015). Concurrently, policy-makers charged with safeguarding food security, stabilizing rural incomes, and taming inflation use these forecasts to calibrate strategic reserve releases, minimum purchase price schemes, and targeted subsidy programs–anticipating upward index movements to deploy stocks that attenuate spikes, or preparing support measures such as direct payments and tax relief in the face of impending price declines to shield producers from revenue shortfalls and forestall indebtedness (Avinash et al. 2024; Xu and Zhang 2023; Wang et al. 2020; Xu 2020; Ouyang et al. 2019; Xu and Zhang 2023; Chen et al. 2021; Xu and Zhang 2023). Moreover, the People's Bank of China and monetary authorities incorporate such price projections into broader inflation forecasting frameworks that inform interest rate and liquidity policies, ensuring responsiveness to the agricultural sector's unique cyclical dynamics without compromising macroeconomic stability (Zhang et al. 2020; Jin and Xu 2025ww; Sari et al. 2024).

In financial markets, composite index forecasts underpin the structuring and valuation of derivative contracts–futures, options, and structured products–that allow participants to transfer price risk to more risk-tolerant investors, thereby deepening liquidity, enhancing price discovery, and attracting domestic and foreign capital to China's commodity trading venues (Vohra et al. 2019; Zhong et al. 2019; Xu and Zhang 2023; Fang et al. 2020; Xu 2017; Kumarmahto et al. 2019; Jin and Xu 2025xx; Sun et al. 2023; Xu and Zhang 2023; Rana et al. 2024; Jin and Xu 2025yy). From an academic standpoint, the continual enhancement of forecasting models for composite indices of the Vegetable Basket Project fosters methodological innovation as scholars benchmark econometric, time-series, and machine-learning techniques–spanning VAR and GARCH frameworks to RNNs and Gaussian processes–against extensive datasets, thereby propelling advances in predictive analytics within applied agricultural economics (Guo et al. 2022; Xu and Zhang 2021; Wang 2023; Xu 2014). Beyond economic and financial spheres, dependable predictions of wholesale indices of the Vegetable Basket Project have profound social welfare implications: by reducing uncertainty around food price inflation, they help protect vulnerable urban and rural households–particularly low-income groups heavily dependent on food expenditures–from destabilizing cost surges, thus preserving purchasing power and alleviating malnutrition risks (Zhang et al. 2023; Zhang and Tang 2024). Transparent and stable price indicators also bolster public confidence in both market-based allocation and governmental oversight, which is critical for maintaining cohesion across China's broad demographic landscape (Hegde et al. 2021; Gowthaman et al. 2023). Environmental objectives likewise benefit, as anticipated price trends can incentivize adoption of precision agriculture technologies–such as satellite monitoring, variable-rate irrigation, and drone applications–that enhance resource-use efficiency; when favorable price windows are forecasted, farmers and enterprises are more inclined to invest in conservation practices that deliver long-term ecological and economic gains, aligning sustainability with productivity targets (Anggraeni et al. 2018; Xu and Zhang 2022; Rani et al. 2022; Xu a; Gu et al. 2022; Xu and Zhang 2023). In the face of climate change–with its attendant shifts in precipitation regimes, temperature extremes, and pest pressures–composite index forecasting of the Vegetable Basket Project provides an early-warning mechanism that identifies vulnerabilities across commodity groups and regions, enabling proactive adaptation measures such as crop diversification, water infrastructure enhancements, and region-specific insurance solutions (Sanusi et al. 2022; Jin and Xu 2025zz). Additionally, amid evolving global trade dynamics and geopolitical uncertainties, robust domestic price projections empower negotiators with quantitative insights into supply elasticity and sensitivity to international shocks, thereby informing evidence-based strategies for tariffs, quotas, and bilateral agreements that protect national interests and support China's growing role in global agricultural markets (Wang et al. 2017; Oktoviany et al. 2021). Finally, the cumulative gains from enhanced forecasting contribute to the broader digital transformation of rural economies–embodied in initiatives like Rural Revitalization and the Belt and Road framework–by underpinning data-driven governance, fostering cross-sector collaboration among government, academia, and private enterprise, and steering China's agricultural sector toward a technology-enabled, inclusive, and sustainable model that sustains rural vitality and national prosperity into the twenty-first century and beyond (Li et al. 2024; Paul and Garai 2021; Xiong et al. 2015; Xu and Zhang 2023; Wu et al. 2017; Geetha et al. 2024; Marfatia et al. 2022; Murugesan et al. 2022).

Despite the well-documented and multifaceted importance of forecasting the composite wholesale price index of the Vegetable Basket Project, a critical examination of the existing methodological landscape reveals a significant gap. Current forecasting approaches, as applied to this specific index, predominantly rely on either traditional econometric time-series models like ARIMA and VAR, or more recent machine learning algorithms such as neural networks and support vector regression. While these methods have provided valuable insights, they collectively exhibit limitations when confronted with the unique characteristics of this long-spanning, policy-sensitive index. Econometric models, with their parametric assumptions of linearity, are inherently challenged by the series' pronounced non-linear dynamics, volatility clustering, and structural breaks induced by major policy shifts, regulatory changes, and macroeconomic upheavals documented over the 2005–2025 period. Conversely, standard machine learning approaches, while adept at capturing non-linearity, often function as "black boxes," failing to provide potential for rigorous, probabilistic uncertainty quantification essential for risk-averse policy-making and financial hedging. Furthermore, the validation strategies for many existing models may not adequately account for the temporal heterogeneity present in such a long series, potentially leading to an overestimation of their true out-of-sample forecasting power. Therefore, a distinct methodological gap exists for a forecasting framework that can simultaneously: (i) flexibly model the complex, non-linear, and non-stationary dynamics inherent to the index without imposing restrictive parametric forms; and (iii) demonstrate robust performance through a validation scheme explicitly designed to test its generalization capability across diverse regulatory and economic regimes. This study directly addresses this gap by introducing a Bayesian-optimized Gaussian Process Regression (GPR) ensemble, a non-parametric probabilistic method whose capacity for structured kernel learning makes it exceptionally well-suited to capture the intricate dynamics and provide reliable forecasts required for this critical economic indicator.

Gaussian process regression (*GPR*) constitutes a powerful Bayesian nonparametric methodology for uncovering latent functional dependencies between predictor variables and associated response measurements. Its theoretical foundations derive from Neal's seminal work demonstrating that infinitely wide neural networks, under Gaussian weight initialization, converge distributionally toward Gaussian processes (Neal 2012). Diverging from traditional parametric regression methods, which enforce restrictive assumptions about specific functional representations, *GPR* directly imposes a Gaussian prior over the infinite-dimensional space of continuous functions by specifying a multivariate normal distribution across latent function evaluations. Intrinsic uncertainty quantification is achieved via integration of an additive Gaussian noise component within the likelihood function, resulting in a joint Gaussian distribution covering both latent functions and noisy observations. Conditioning this joint prior on empirical training data generates a posterior distribution over function values at unexamined locations, featuring analytically tractable mean and covariance expressions. This analytical tractability avoids computationally demanding sampling schemes typically required within comprehensive Bayesian inference frameworks (Williams and Rasmussen 1995; Neal 1997). Early empirical assessments revealed that *GPR* often exceeds the predictive precision of radial-basis-function neural networks when target phenomena exhibit stationary properties masked by observational noise, especially in contexts involving complex nonlinear interactions (Brahim-Belhouari and Vesin 2001). Subsequent scholarly developments have significantly broadened the repertoire of available

covariance kernels. While the conventional squared-exponential kernel enforces stringent smoothness constraints, the Matern family permits tunable differentiability degrees. Kernels with periodic structure explicitly incorporate cyclical patterns, whereas rational-quadratic kernels efficiently characterize processes possessing multiple distinct characteristic scales. Moreover, composite covariance architectures–formed through additive or multiplicative amalgamations of elementary kernels–facilitate representation of nonstationary behavioral regimes without compromising analytical tractability for essential computations. These critical computations encompass marginal likelihood assessment and covariance matrix inversion (Brahim-Belhouari and Bermak 2004).

Despite substantial expressive versatility, a basic *GPR* implementation suffers from $\mathcal{O}(n^3)$ computational burden arising from inversion of the complete $n \times n$ covariance matrix, where n denotes training sample size. To alleviate this limitation, hyperparameter optimization may advance through gradient-based maximization of the log marginal likelihood, or fully Bayesian approaches can integrate over parameters using Markov chain Monte Carlo methodologies. More recently, sparse approximation techniques have risen to prominence: inducing-point methods select a condensed subset of representative pseudo-inputs to approximate the full covariance, while variational inference frameworks impose low-rank factorizations upon the covariance structure. These advances substantially reduce computational overhead, enabling *GPR* application to extensive problems across climate modeling, financial econometrics, and intricate systems identification. Extending these methodological improvements, the present study proposes an ensemble framework amalgamating multiple specialized *GPR* models. Each component model utilizes a dedicated covariance kernel crafted to capture particular temporal attributes–such as extended dependencies, seasonal cyclicity, or sudden shifts within trend and volatility phases. Aggregating posterior predictions from these heterogeneous kernel specifications reduces overreliance on any single assumption while improving responsiveness to structural transitions common within economic time series. For example, synergistically merging kernels adept at modeling gradual secular trends with those optimized for identifying sharp discontinuities allows the ensemble to represent both slowly evolving market tendencies and abrupt policy-driven disturbances. Notably, the analytically derived posterior covariance inherently supplies well-calibrated credible intervals, obviating dependence on bootstrap resampling or similar techniques for quantifying predictive uncertainty (Williams and Rasmussen 1995; Neal 1997). To further enhance interpretability and modeling flexibility, hierarchical kernel constructions decompose overall covariance into semantically meaningful strata. This stratification separates long-term trend components from seasonal fluctuations, captures spatial interdependencies among distinct geographical zones, and incorporates exogenous predictors such as macroeconomic indices or weather variables. Such structured kernels establish interpretable linkages between individual covariance elements and the underlying variability sources they represent, while retaining the expressive power necessary for characterizing the multifaceted dynamics of composite agricultural price indicators of the Vegetable Basket Project. These dynamics are shaped by global supply–demand imbalances, climatic disruptions, and regulatory changes. Concurrently, *GPR*'s methodological adaptability–encompassing its capacity for exact posterior inference, flexibility in encoding diverse spatiotemporal dependencies, and compatibility with computationally efficient approximations–positions it as an exceptionally relevant instrument for projecting complex economic metrics and elucidating fundamental drivers of price dynamics.

Capitalizing on the intrinsic adaptability of Gaussian process regression (*GPR*), a non-parametric Bayesian statistical paradigm, we establish a Gaussian process prior distribution characterizing the latent functional mechanism governing temporal price evolution. Deviating from constraining algebraic presuppositions, our methodology algorithmically deduces covariance kernel configuration parameters directly from historical pricing records. Within this probabilistic framework, essential parameters–encompassing characteristic length scales dictating temporal dependency horizons and noise-variance components quantifying measurement uncertainty–are ascertained via data-driven inference procedures instead of heuristic assignment. These hyperparameters experience optimization through marginal likelihood maximization utilizing Bayesian optimization techniques, systematically obviating subjective manual calibration. Consequently, the process exhaustively navigates high-dimensional posterior surfaces to pinpoint kernel configurations optimizing predictive fidelity. To counteract overfitting and guarantee dependable generalization capability assessment, a stratified k-fold cross-validation protocol is deployed. This technique segments the entire dataset into k mutually exclusive partitions, each conserving stochastic properties intrinsic to the complete series. Iteratively, every partition functions as validation data while the residual $k-1$ partitions form training data, ensuring all observations engage in both model development and evaluation phases. This approach yields nearly impartial out-of-sample prediction error approximations, validating that performance metrics authentically represent extrapolative capacity rather than in-sample fitting.

The present work covers the period of September 27, 2005 to April 10, 2025 with February 17, 2022–April 10, 2025 serving as the out-of-sample testing phase, which extends the GPR method by selecting a significantly longer out-of-sample time horizon than conventional ones. During stringent out-of-sample assessment employing wholesale agricultural commodity composite price indices of the Vegetable Basket Project spanning February 17, 2022, to April 10, 2025, our optimized ensemble yielded these performance indicators: a relative root mean square error (*RRMSE*) of 0.2040%, a root mean square error (*RMSE*) of 0.4841, a mean absolute error (*MAE*) of 0.3430, and a correlation coefficient (*CC*) of 0.99929. These aggregate metrics attest to the model's proficiency in tracking actual price pathways with remarkable precision and minimal deviation. Our *GPR* architecture enables effortless integration within hybrid analytical workflows merging established econometric methodologies–such as Vector Error-Correction Models (VECMs) or Autoregressive Integrated Moving Average (ARIMA) systems–with supplementary machine-learning approaches like gradient-boosted decision trees or support-vector machines. Illustratively, a VECM might capture long-run equilibrium relationships, while the Gaussian process characterizes residual temporal dependencies to improve composite forecast precision. Conversely, an ARIMA specification could extract deterministic trends, permitting the *GP* to concentrate exclusively on stochastic volatility residuals. Such modular designs bolster resilience against sudden market structural discontinuities. Additionally, the framework supports conditional stochastic simulation functionality through posterior distribution sampling. Analysts can consequently produce counterfactual future price trajectories under alternative macroeconomic or regulatory scenarios, scrutinize outcome distributions, pinpoint adverse tail-risk occurrences, and compute exceedance probabilities for critical policy thresholds. These projections are vital for strategic contingency planning amid uncertainty, elucidating transmission pathways for exogenous perturbations through pricing architectures.

To ensure methodological transparency and computational reproducibility, the entire analytical sequence–including raw data preprocessing, standardization procedures, kernel hyperparameter optimization, and cross-validation partitioning–is executed within thoroughly documented computational scripts. All intermediate operations (e.g., covariance matrix calculations, cross-validation subset generation, hyperparameter search space definition) are explicitly delineated, facilitating straightforward application to different markets or refreshed datasets absent substantial alteration. By synthesizing adaptive Bayesian learning with established wholesale agricultural commodity composite price analytics for the Vegetable Basket Project, our technique swiftly identifies regime transitions while furnishing rigorous probabilistic uncertainty quantification. Ultimately, this ensemble-Gaussian process regression framework establishes a scalable, interpretable, and robust modeling foundation for empirically grounded policy development. Through its amalgamation of probabilistic forecasting, modular hybrid assimilation, and scenario-based simulation capacities, the methodology empowers decision-makers to evaluate central projections and distributional tail risks within agricultural commodity markets, thereby cultivating more transparent and resilient macroeconomic governance structures exhibiting enhanced resistance to systemic disruptions.

To the best of our knowledge, this research constitutes a distinct methodological contribution by presenting the implementation of a Bayesian-optimized Gaussian process regression ensemble, specifically tailored with a cross-validation scheme, for forecasting the composite wholesale agricultural commodity price index of China's Vegetable Basket Project. While the broader application of machine learning, including neural networks and support vector machines, to agricultural commodity price forecasting is an active and well-documented field, and while GPR itself has been explored in adjacent financial and energy econometrics contexts, its potential remains largely untapped for high-frequency, composite agricultural indices characterized by pronounced policy-induced structural breaks. Previous studies often rely on standard validation techniques that may not adequately account for the temporal heterogeneity and regime shifts inherent in such a long-span economic series. Our approach's novelty lies not in the use of GPR in vacuo, but in the synthesis of a sophisticated, probabilistic nonparametric model within a rigorous validation architecture explicitly designed for economic policy analysis. This integrated framework–combining heterogeneous kernel specification, hyperparameter calibration via the Expected Improvement Per Second Plus (EIPSP) acquisition function, and a strategy that ensures robustness across diverse regulatory epochs–enables a uniquely resilient and uncertainty-aware forecasting system. Therefore, we narrow our claim of novelty to this specific, domain-focused integration: the development and validation of a GPR ensemble, optimized within a Bayesian framework and validated against the partition of training and testing phases, for the particular challenge of forecasting the Vegetable Basket Project index, a critical barometer of China's food price stability. This methodology provides a new, powerful tool for a problem domain where forecasting accuracy is directly tethered to national economic planning and social welfare outcomes. We have also conducted benchmark analysis against other machine learning models and traditional time-series econometric models, which demonstrate the advantage of our model for the forecasting purpose for the Vegetable Basket Project index.

The important problem this study confronts is twofold. Empirically, the index exhibits a complex mixture of non-linear dynamics, long-memory volatility persistence, and–most

critically–abrupt structural breaks induced by two decades of policy interventions, regulatory overhauls, and macroeconomic transitions. Methodologically, existing forecasting approaches have proven inadequate for this specific task: classical econometric models such as ARIMA and VAR could be misspecified for the index's non-stationary, policy-driven regime shifts, while standard machine learning algorithms such as LSTM and SVR, although flexible, lack a systematic mechanism for hyperparameter calibration that ensures robust out-of-sample generalization across heterogeneous regulatory epochs. This dual inadequacy remains unresolved in the literature, as previous studies have neither explicitly modeled the index's susceptibility to policy-induced discontinuities nor validated their forecasts against a testing protocol that spans multiple distinct structural regimes. Our framework directly addresses this gap by synergistically combining three elements specifically chosen to tackle these challenges: (i) a kernel composition within the GPR that can simultaneously represent smooth trend evolution and abrupt structural transitions; (ii) a Bayesian optimization routine that automatically calibrates model complexity to the index's unique temporal signatures without subjective intervention; and (iii) a cross-validation architecture that deliberately exposes the model to the full spectrum of regulatory environments encoded in the dataset, thereby ensuring out-of-sample robustness. Our contribution is not the invention of Gaussian process regression itself, but rather the demonstration that a carefully architected GPR ensemble, optimized within a rigorous Bayesian validation framework explicitly designed for policy-sensitive, structurally unstable time series, can resolve the longstanding forecasting challenges associated with this critical economic barometer. This provides policymakers and market participants with a powerful, empirically validated tool for navigating an index whose behavior is fundamentally shaped by the intersection of market forces and state intervention–a context where conventional forecasting paradigms have consistently fallen short.

2 Literature review

Econometric methods applied to time-series data form a core methodological paradigm for examining price formation processes and evolutionary pathways within commodity and financial sectors (Makridakis et al. 2022; Jin and Xu 2025aaa; Illankoon and Kumara 2020; Xu and Zhang 2024; Arunraj and Ahrens 2015; Xu and Zhang 2023; Liu and Yu 2019; Ji et al. 2022; Arunraj et al. 2016; Xu and Zhang 2023; Prakash and Sumaiya Farzana 2019). These quantitative tools provide structured analytical scaffolding capable of explicitly representing temporal interdependencies, detecting structural breaks, and incorporating stochastic trends to model complex dynamic behaviors, volatility clustering episodes, and persistent effects inherent in such systems (Xu and Zhang 2023; Sreerag and Shanmugam 2023; Xu 2018; Divisekara et al. 2020; Xu 2015; Navaratnalingam et al. 2020; Xu b). The methodological versatility embodied by these techniques has received extensive empirical validation across recent decades through rigorous studies of agricultural supply chains. Academic research has comprehensively chronicled the transmission mechanisms through which upstream disturbances–emanating from production areas and propagating along processing networks toward retail destinations–affect end-consumer price stability (Babu and Mallikharjuna Rao 2022; Jin and Xu 2025a; Alhussan et al. 2024; Jin and Xu 2025b; Mayer et al. 1991; Jin and Xu 2025c; Dhaka et al. 2023; Jin and Xu 2025d; Kim et al. 2020). This scholarly corpus often utilizes advanced autoregressive methods integrated with

cointegration analysis to map interregional commercial connections, measure spatial price convergence among geographically separated trading centers, and evaluate pricing structure transformations across value chain stages arising from supply-chain efficiency modifications. Investigations clarify how primary disruptions–caused by climatic events, input cost variations, or regulatory shifts–disseminate through logistical and processing systems, ultimately affecting regional market integration and consumer welfare outcomes. Comparable methodological refinements appear in energy market research, where academics decompose volatility patterns, identify regime transitions (e.g., changes between stable and turbulent states), and isolate contemporaneous linkages binding conventional fossil fuel benchmarks with emerging carbon pricing mechanisms (Olorunnipa and Florkowski 1993; Xu and Zhang 2023; Kostaridou et al. 2024; Xu and Zhang 2023a). Employing regime-switching volatility frameworks, studies illustrate how abrupt crude oil or natural gas price movements permeate emissions trading schemes, uncovering bidirectional dependencies essential for comprehending risk transmission pathways and predicting spillover likelihoods. Similar applications reach financial instruments during heightened uncertainty periods, integrating stochastic volatility specifications, evaluating risk-adjusted strategy performance, and quantifying co-movements linking equities, precious metals, and derivative contracts. These scholarly pursuits enhance understanding of cross-asset connections, demonstrating, for instance, gold or platinum's potential hedging utility against equity declines or raw material price fluctuations' impact on systemic financial risk indicators and portfolio diversification approaches (Xu and Zhang 2023b; An et al. 2020; Jin and Xu 2025e; Benz and Trück 2009; An and Mikhaylov 2020; Jin and Xu 2025f; Benchimol and El-Shagi 2020; An et al. 2024; Jin and Xu 2025g; Dinçer et al. 2024; Paolella and Taschini 2008).

Within specialized arenas like emissions trading, equilibrium-based temporal frameworks elucidate dynamic interactions between constrained allowance availability and evolving abatement necessities. These models characterize how regulatory revisions–including cap adjustments or offset inclusion–modify marginal compliance expenses and govern permit price discovery across varied policy environments; such examinations reveal how anticipatory actions driven by expected regulatory stringency can incentivize strategic trading, alter inventory positions, and subsequently influence allowance price paths (Gubanova et al. 2005; Mengdi and Yong 2018; Xu and Zhang 2023, 2024). Practitioners routinely encounter substantial methodological obstacles originating from nonstationarity within the first and second moments of price sequences–conditions potentially compromising parameter estimation consistency and inferential validity. This reality demands thorough preliminary diagnostic examination, involving unit-root tests, precise cointegration rank determination, and appropriate variance-stabilizing transformations to obtain statistically sound and economically interpretable coefficient estimates (Kuhns et al. 2015; Jin et al. 2025; Bessler 1982; Xu and Zhang 2023). Overlooking these diagnostic steps risks producing spurious regression outcomes, coefficient distortions, and incorrect policy conclusions, particularly when modeling volatility oscillating between calm and chaotic phases. This complexity has catalyzed methodological progress, including composite likelihood estimators, Markov-switching variance formulations, and generalized autoregressive conditional heteroskedasticity (GARCH) model extensions. These enhanced approaches accommodate sudden variance alterations–frequently induced by exogenous incidents like geopolitical conflicts, severe weather, or major policy initiatives–allowing researchers to capture volatility spikes and enhance predictive precision across diverse market conditions (Kupfer-

schmidt and Kupferschmidt 2024; Jin and Xu 2025h; Benschopa and López Cabreraa 2014; Xu and Zhang 2021).

Moreover, endeavors to represent nonlinear interdependencies among macroeconomic variables–encompassing GDP trajectories, interest rate movements, and sectoral demand elasticities–have spurred nonparametric/semiparametric modeling innovations, threshold autoregressive systems, and hybrid architectures merging machine-learning algorithms to detect regime-dependent nonlinearities. These techniques excel at identifying structural thresholds, such as junctures where monetary policy changes transform commodity demand pressures, offering superior flexibility where relationship strength or direction evolves alongside shifting economic contexts (Wu et al. 2017; Xu and Zhang 2023; Jin and Xu 2025i; Xu and Zhang 2024). Classical autoregressive integrated moving average (ARIMA) specifications retain relevance owing to their parsimonious inclusion of autoregressive terms, differencing operations, and moving-average components, effectively isolating cyclical and seasonal variations within agricultural and energy prices (Liu et al. 2021; Xu and Zhang 2024; Zhou et al. 2022; Xu and Zhang 2022; Xu et al. 2024; Xu and Zhang 2022; Kling and Bessler 1985; Jin and Xu 2025j; Bessler 1982; Brandt and Bessler 1981; Jin and Xu 2025k; McIntosh and Bessler 1988; Bessler 1990; Xu and Zhang 2024; Jin and Xu 2025l; Xu 2017, 2020). Augmenting ARIMA with error-correction constraints imposing long-run equilibria and state-dependent coefficients sensitive to policy alterations bolsters its resilience against supply interruptions, regulatory modifications, and global demand shifts. Vector autoregressive (VAR) systems prove essential for disentangling interconnected dynamics in multivariate contexts, permitting concurrent estimation of reciprocal influences (Bessler and Brandt 1992; Jin and Xu 2025m; Chen and Bessler 1987; Xu 2018; Zhao et al. 2020). They enable impulse-response analyses tracking shock propagation (e.g., crude oil price surges affecting commodity complexes and financial indices) and facilitate Granger causality testing employing spatially disaggregated or high-frequency data–capabilities critical for regulators anticipating cross-market contagion stemming from sector-specific actions. Contemporary progress involves implementing Bayesian shrinkage priors to reduce overparameterization and utilizing time-varying coefficient matrices to depict evolving structural linkages, tackling estimation difficulties spanning cryptocurrency valuations to post-recession recoveries. Bayesian approaches encourage parsimony and predictive stability by regularizing insignificant coefficients, while time-varying coefficients adjust dynamically to sentiment or policy shifts (Paolella and Taschini 2008; Moon et al. 2019; Yan et al. 2020; Dutta et al. 2019; Xu 2017, 2019).

For cointegrated systems, vector error-correction models (VECMs) integrate long-run equilibrium constraints with short-term adjustment mechanisms, enabling detailed scrutiny of spatial price convergence in agricultural goods and the dynamic connection between emissions allowance costs and energy prices (Bessler et al. 2003; Jin et al. 2025; Yang and Leatham 1998; Xu 2015; Jin and Xu 2025n; Xu and Zhang 2023a; Zhu et al. 2019; Jin and Xu 2024a). By embedding these equilibrium associations, VECMs guarantee temporary disequilibria correct progressively, proving vital for analyzing intrinsically interconnected variables. Incorporating threshold effects and asymmetric adjustment processes further increases adaptability, reflecting differential reversion speeds dependent on deviation direction, thereby capturing asymmetric risk perceptions and adjustment frictions. Interdisciplinary efforts increasingly combine econometric rigor with machine-learning adaptability, creating unified analytical paradigms. Illustrations include Markov-switching

GARCH frameworks capturing regime-specific volatility persistence, threshold autoregressive systems tailored for nonlinear discontinuities, and combined ARIMA–neural network or ARIMA–support vector machine constructs harnessing nonparametric pattern recognition for superior predictive capability (Zhao et al. 2018; Xu and Zhang 2023, 2022, 2023). Deployed for carbon derivatives, critical mineral sectors, geopolitically sensitive commodities, and large-scale shock transmission analyses, these composite methodologies attain enhanced forecasting accuracy amid nonlinear dynamics or high-dimensional interactions. Simultaneously, methodological scholarship has established stringent validation protocols incorporating recursive estimations, cross-validation procedures, and forecast-error decompositions for objective comparative evaluation. This guarantees consistent performance assessment across temporal horizons, reducing overfitting dangers and improving generalizability. Researchers have further explored market informational efficiency under differing structural states and investigated welfare implications of commodity speculation, determining whether price movements primarily reflect fundamental supply-demand drivers versus transient speculative forces–yielding crucial insights for regulatory architecture and surveillance mechanisms (Bessler and Brandt 1981; Jin and Xu 2025o; Bessler and Hopkins 1986; Jin and Xu 2025p; Bessler and Babula 1987; Jin and Xu 2024b; Brandt and Bessler 1982; Jin and Xu 2025r; Brandt and Bessler 1983; Jin and Xu 2024c; Chen and Bessler 1990; Xu and Zhang 2021; Yang and Awokuse 2003; Jin and Xu 2025s; Luckow et al. 2015; Xu 2019; Jin and Xu 2024d; Xu 2019; Benschopa and López Cabreraa 2014). Collectively, this vast literature outlines a dynamic academic domain where continual methodological enhancement, evolving regulatory frameworks, and escalating sustainability demands converge to strengthen the robustness of price time-series forecasting. As market interconnectedness deepens and data accessibility grows, persistent advances in model specification, estimation methodologies, and validation criteria remain crucial for navigating the complexities of progressively interdependent market ecosystems.

Computational economic simulation integrated with data-intensive analytical approaches has instigated a profound shift in economic modeling paradigms, significantly accelerated by machine-learning algorithms designed to predict price movements in varied market environments. Scholarly research increasingly prioritizes utilizing high-dimensional computational frameworks to illuminate fundamental price discovery processes and optimize policy measures (Petropoulos et al. 2022; Jin and Xu 2025t; Zhang et al. 2024; Jin and Xu 2025u; Yang 2024; Fatima and Uddin 2022; Jin and Xu 2025v). A pivotal enabler of this methodological transition involves the extensive availability of fine-grained market transaction records, which document intricate microstructure features–such as bid-offer price differentials, trade execution details, and order book depth–permitting unprecedented scrutiny of intra-period pricing dynamics. These continuous, high-frequency data feeds allow detection of nuanced fluctuations stemming from algorithmic trading operations, automated order placements, and ephemeral liquidity imbalances. Simultaneously, the widespread adoption of scalable cloud-computing platforms and distributed processing architectures has substantially reduced both financial and technical barriers associated with handling voluminous, near-real-time information streams. This enables effective acquisition, retention, and computational processing of terabyte-level transaction repositories without dependence on specialized on-premises infrastructure (Makridakis et al. 2022; Xu and Zhang 2024; Adekoya 2021; Xu and Zhang 2022; Zhang et al. 2025; Jin and Xu 2025x; Nadirgil 2023; Xu and Zhang 2022; Wang et al. 2023; Jin and Xu 2024e). Consequently, contemporary compara-

tive analyses systematically evaluate multiple algorithmic strategies–including supervised techniques like support vector regression (SVR) and multilayer perceptrons, unsupervised clustering for latent state identification, and ensemble methods integrating base predictors–each carefully optimized to detect abrupt structural changes, represent intricate volatility regimes, and assimilate disruptions from regulatory amendments or policy developments (Petropoulos and Spiliotis 2021; Jin and Xu 2024f; Alade et al. 2021; Jin and Xu 2024g; Alade et al. 2022; Jin and Xu 2024h; Alade et al. 2020; Jin and Xu 2025y; Adewumi et al. 2016).

Empirical assessments consistently reveal these adaptive systems outperform conventional econometric specifications in predictive precision–especially within contexts characterized by pronounced information asymmetries, heterogeneous agent interactions, and intermittent regime transitions (Jin and Xu 2025z; Wang et al. 2024; Xu and Zhang 2023; Zhang and Deng 2025; Sayed et al. 2024; Jin and Xu 2024i; Lei et al. 2022; Xu and Zhang 2023; Zhang and Wen 2022; Jin and Xu 2024j). Deep feedforward neural networks find broad application for modeling localized price indicator variations, leveraging their hierarchical nonlinear feature extraction capacities and proficiency for uncovering latent interdependencies potentially obscured within linear specifications. Concurrently, recurrent neural configurations, particularly long short-term memory (LSTM) designs, demonstrate outstanding competence for capturing persistent temporal autocorrelations and volatility clustering intrinsic to financial series, attributable to their inherent long-memory characteristics and nonlinear shock propagation pathways (Xu and Zhang 2022; Sun et al. 2016; Jin and Xu 2024k; Huang et al. 2021; Jin and Xu 2024l; Zhang et al. 2018; Xu and Zhang 2023; Han et al. 2015; Xu and Zhang 2025). These deep-learning methods display marked effectiveness when managing complex temporal relationships and market activities reflecting both discrete catalysts–such as regulatory announcements–and prolonged, state-dependent trajectories. Stringent validation across diverse industrial domains, encompassing strategic metals exchanges, bulk commodity platforms, and international carbon markets, substantiates such architectures maintain reliable forecasting efficacy despite nonstationary inputs, considerable noise contamination, and data paucity, frequently surpassing sophisticated nonlinear econometric alternatives (Mori and Jiang 2008; Jin and Xu 2024m; Mori and Jiang 2008; Xu and Zhang 2023; Jaramillo-Morán and García-García 2019; Yahşi et al. 2019; Xu et al. 2020; Lu et al. 2020; Fan et al. 2015; Liu and Shen 2020; Han et al. 2019; Jaramillo-Morán et al. 2021; Jiang and Wu 2015; Ameyaw et al. 2019; Hong et al. 2017; García and Jaramillo-Morán 2020; Li et al. 2020; Atsalakis 2016).

Beyond neural paradigms, researchers have customized various algorithmic families to address distinctive attributes of price paths, frequently involving regulatory intricacy and ambiguous information dissemination. For example, extreme learning machines (ELMs) achieve swift convergence through randomized hidden-layer parameterization combined with analytical output weight determination, attaining an advantageous equilibrium between predictive fidelity and computational efficiency (Zhou and Wang 2021; Hao and Tian 2020; Zhao et al. 2021; Li et al. 2022; Sun and Duan 2019; Ren and Long 2021; Zhou and Chen 2021; Zhang et al. 2019; Xu et al. 2020; Huang and He 2020; Chai et al. 2021; Wei et al. 2018; Zhou et al. 2019; Hao et al. 2020; Jianwei et al. 2021; Chai et al. 2020; Zhu and Chevallier 2017; Lu et al. 2020; Sun and Wang 2020; Zhu et al. 2021, 2017; Chen et al. 2021; Jiang and Wu 2015; Li et al. 2021; Qin et al. 2020; Zhu et al. 2018; Abdi and Taghipour 2019; Jaramillo-Morán et al. 2021; Hong et al. 2017). Support-vector regression systems uti-

lize kernel functions to represent complex nonlinear relationships within high-dimensional predictor spaces, preserving robust generalization capabilities. Hybridized fuzzy-stochastic techniques amalgamate fuzzy logic for handling epistemic uncertainty with stochastic elements for depicting aleatory variability, collectively addressing imprecision and randomness inherent in market fundamentals. Gradient-boosting decision tree ensembles, including XGBoost and LightGBM, progressively refine forecasts by concentrating on residual errors from earlier stages, facilitating recognition of subtle nonlinear interactions among price drivers. Composite frameworks combining random forests, Bayesian belief networks, and adaptive weighting schemes further enhance generalizability by exploiting complementary learner advantages and reducing overfitting; an integrated architecture might employ random forests for feature relevance quantification, Bayesian networks for state transition likelihood estimation, and adaptive layers to dynamically rebalance submodel contributions reacting to prevailing market conditions–such as regulatory revisions or volume anomalies–thereby increasing resilience.

Despite the evident predictive enhancements afforded by these sophisticated learning approaches, practitioners confront persistent challenges regarding model transparency and compliance with regulatory requirements. Numerous stakeholders and policymakers exhibit reluctance to rely on non-transparent models with obscure internal decision mechanisms, particularly when outputs inform crucial judgments–such as setting auction reserve thresholds or distributing emissions permits. Nevertheless, practical implementations exhibit tangible benefits: neural-network-augmented demand forecasting enables finer calibration of auction price minima, diminishing volatility and enhancing market fluidity, while ensemble-based scenario simulation instruments assist regulators in anticipating circumstances where supply gluts or unforeseen policy shifts could endanger sustainability goals (Yang et al. 2020; Xu 2018; Zhang et al. 2019; Ren and Long 2021; Xu and Zhang 2023a; Wang et al. 2021). These methodological advances, by providing superior forecasting accuracy and responsiveness to evolving market realities, deliver significant utility for projecting wholesale agricultural commodity composite price indices of the Vegetable Basket Project. Accurate characterization of abrupt cost oscillations and supply-chain interruptions within this agricultural sphere is vital for formulating effective economic strategies and preserving macroeconomic equilibrium. The capacity to handle high-velocity data, represent nonlinear discontinuities, and dynamically adjust to foundational structural shifts establishes machine-learning architectures as essential instruments for navigating contemporary price dynamics, ultimately strengthening both market functionality and regulatory responsiveness.

3 Data

Over the last twenty years, daily observations of China's wholesale agricultural commodity composite price index of the Vegetable Basket Project (spanning September 27, 2005, to April 10, 2025), sourced from Ministry of Agriculture and Rural Affairs of the People's Republic of China, have experienced significant economic transformations. These structural shifts have induced marked nonlinear dynamics and episodic structural disruptions, violating stationarity assumptions underpinning conventional linear autoregressive moving-average (ARMA) techniques. Initial graphical examination of temporal price evolution and distributional characteristics (Fig. 1) discloses pronounced volatility persistence along-

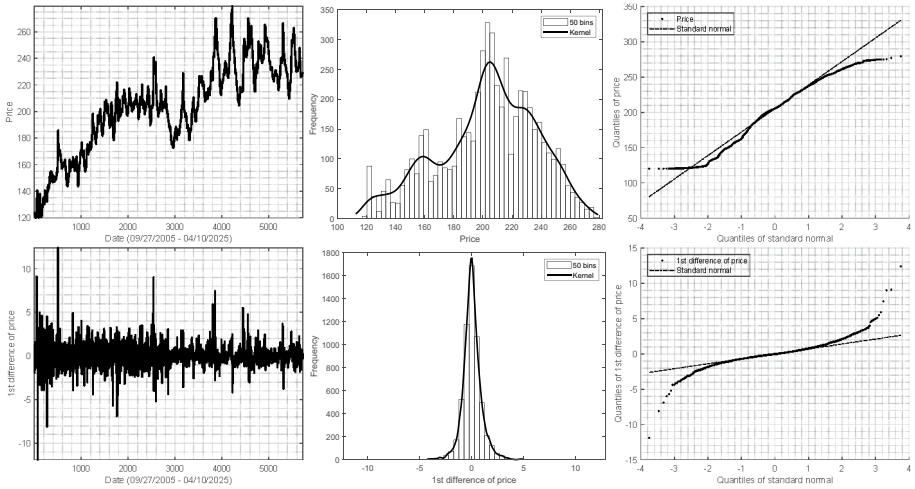


Fig. 1 Daily composite wholesale agricultural commodity price indices of the Vegetable Basket Project, 09/27/2005–04/10/2025

side substantial tail-risk manifestations. Supplementary descriptive statistics presented in Table 1 (including mean, median, standard deviation, skewness, kurtosis) reveal considerable deviations from Gaussian distributional attributes. Furthermore, formal normality assessments–incorporating Jarque–Bera and Anderson–Darling diagnostics–consistently reject the Gaussian null hypothesis at conventional significance thresholds, confirming standard ARMA frameworks' limitations in representing higher-order moments and heteroskedastic patterns within this dataset (Xu and Zhang 2021; Jin and Xu 2025bb; Xu and Zhang 2024; Jin and Xu 2025cc; Xu and Zhang 2023; Jin and Xu 2025dd). Addressing these empirical intricacies, this research positions the wholesale agricultural commodity composite price index series of the Vegetable Basket Project within an enhanced financial-econometric framework, categorizing prospective methodological innovations into four principal domains. Firstly, generalized autoregressive conditional heteroskedasticity (GARCH) extensions–such as exponential GARCH (EGARCH) and fractionally integrated GARCH (FIGARCH)–explicitly characterize time-varying conditional variance and asymmetric innovation responses: EGARCH accommodates leverage effects via differential volatility impacts between positive and negative shocks, while FIGARCH utilizes fractional differencing operators to model long-memory persistence within variance clustering (Liu et al. 2021; Brandt and Bessler 1981). Secondly, regime-switching architectures, including Markov-switching and smooth-transition autoregressive formulations, furnish stochastic mechanisms for identifying latent structural breaks–whether arising from policy interventions or macroeconomic volatility–through probabilistic transitions among discrete states featuring distinct mean-variance regimes. Thirdly, kernel-based nonparametric techniques–exemplified by Nadaraya–Watson estimators and local-polynomial smoothing–circumvent rigid functional presumptions via data-adaptive weighting schemes capturing complex nonlinear dependencies among lagged prices and economic covariates; sophisticated bandwidth selection methodologies or adaptive kernel configurations mediate the bias–variance trade-off, preserving localized features while avoiding excessive smoothing.

Table 1 Data summary of daily composite wholesale agricultural commodity price indices of the Vegetable Basket Project, 09/27/2005–04/10/2025

Series	Minimum	1st Percentile	5th Percentile	Mean	Median	Standard Deviation	95th Percentile	99th Percentile	Maximum	Skewness	Kurtosis	Jarque–Bera	Anderson–Darling
Price index	119.9000	122.3000	141.6800	202.9124	205.4000	33.7639	253.5385	267.0641	279.4098	−0.3820	2.6133	<0.001	<0.0005
First differences	−11.9000	−2.4000	−1.3030	0.0183	0.0000	0.9723	1.5721	2.7961	12.4000	0.4146	18.4978	<0.001	<0.0005

Fourthly, Gaussian process regression (*GPR*) represents a fully Bayesian nonparametric approach defining covariance functions parameterized by length-scale and signal variance hyperparameters, generating analytically tractable posterior moments while intrinsically quantifying predictive uncertainty, thereby naturally accommodating heteroskedasticity and leptokurtosis (Williams and Rasmussen 1995; Jin and Xu 2025ee; Neal 1997; Jin and Xu 2025ff). Collectively, these four research trajectories–conditional heteroskedastic modeling (GARCH), latent regime dynamics (Markov-switching), flexible nonlinear mapping (kernel regression), and probabilistic forecasting (*GPR*)–form a synergistic analytical toolkit for characterizing the multifaceted stochastic behavior of wholesale agricultural commodity composite price indicators of the Vegetable Basket Project. A systematic comparative assessment of their predictive efficacy offers prospects not only for improved forecasting precision but also deeper theoretical comprehension of price-discovery mechanisms and more resilient, empirically informed directives for policy design and risk management within China's agricultural commodity markets. This comparative analysis further illuminates transmission channels for exogenous disturbances and aids development of early-alert systems for potential market instability, ultimately enhancing sectoral resilience and advancing food security governance frameworks.

Empirical investigations widely substantiate that economic and financial temporal sequences commonly exhibit complex nonlinear patterns exceeding the descriptive scope of conventional linear modeling frameworks. To assess whether China's wholesale agricultural commodity composite price index series of the Vegetable Basket Project similarly displays higher-order dependencies transcending elementary linear autocorrelation architectures, we administer the Brock–Dechert–Scheinkman (BDS) diagnostic developed by Broock et al. (Brock et al. 1996). This distribution-independent test metric quantifies discrepancies between empirically observed correlation integrals–calculated across multiple embedding dimensions (e.g., dimensions 2–5) and neighborhood radii (ϵ)–and their theoretical expectations under an independent and identically distributed null hypothesis. Implementation utilizing daily aggregate agricultural commodity pricing indicators of the Vegetable Basket Project yields consistently statistically significant BDS statistics across every dimension-radius pairing, demonstrating negligible p-values. This outcome conclusively refutes both strict independence postulates and linear autoregressive model formulations, verifying enduring temporal dependencies and intricate nonlinear dynamics. These empirical observations align with theoretical propositions whereby intermittent policy interventions and transient market disequilibria instigate abrupt regime transitions and persistent volatility clustering phenomena. Given these diagnostic determinations, resilient forecasting methodologies necessitate adaptable approximation of unspecified functional relationships coupled with rigorous predictive uncertainty quantification. Gaussian process regression (*GPR*) constitutes an intrinsically appropriate paradigm by instituting a Bayesian prior conceptualized as a Gaussian stochastic process governing the latent price-generation mechanism. This strategy avoids restrictive parametric presumptions while defining a joint multivariate Gaussian distribution across latent function evaluations and observational disturbances. The methodology's foundation resides in covariance function specification: fundamental kernels including squared-exponential (radial basis) or Matern families characterize smooth, monoscale variations. Composite kernels–formed through additive or multiplicative amalgamations of elemental kernels–concurrently model heterogeneous temporal phenomena such as periodic seasonality and structural discontinuities. Illustratively, incorporating peri-

odic constituents accommodates systematic seasonal or industrial cycles, while integrating mean-reverting covariance frameworks captures reversion dynamics typical of commodity markets. Through deliberate kernel synthesis, *GPR* coherently unifies gradual trend evolution and abrupt structural modifications within a consistent probabilistic architecture. Hyperparameter optimization–encompassing kernel length-scales (modulating dependency horizons), signal-variance coefficients (governing output scale), and noise-variance parameters (quantifying measurement imprecision)–advances via log marginal likelihood maximization conducted through Bayesian optimization. This approach balances global exploration of hyperparameter configurations against local refinement employing acquisition functions such as expected improvement. By integrating the BDS test's nonparametric identification of nonlinear dependence with *GPR*'s kernel-based uncertainty quantification, we establish a consolidated forecasting system capable of representing smooth trajectories, structural breaks, and volatility clustering within a unified probabilistic framework. The resulting predictive distributions display comprehensive probabilistic calibration, providing not merely point forecasts but also credible intervals indispensable for risk-aware decision-making. Consequently, policymakers and market participants acquire operationally pertinent insights regarding evolving wholesale agricultural commodity composite price dynamics of the Vegetable Basket Project, enabling improved resilience and empirically anchored strategies amid structural transformation and economic uncertainty. This architecture additionally facilitates probabilistic scenario simulation and tail risk assessment essential for robust agricultural policy formulation amidst volatile market conditions.

While the presence of non-linearity, volatility persistence, and structural shifts is a recognized characteristic of many financial and commodity price series, the specific manifestation of these features within China's composite wholesale agricultural price index for the Vegetable Basket Project presents a unique and unresolved challenge for the existing forecasting literature. The prolonged temporal horizon of this index, spanning two decades of profound economic transition, has encoded a series of policy-driven regime changes–from agricultural subsidy reforms and trade liberalization episodes to strategic reserve interventions–that collectively generate a data-generating process far more complex than those typically encountered in shorter or more institutionally stable price series. Previous efforts to generate point forecasts for China's composite agricultural price indices have predominantly relied on two methodological streams, each exhibiting critical inadequacies when confronted with this specific index. The first stream, grounded in classical econometrics, employs univariate (e.g., ARIMA) or multivariate (e.g., VAR) linear frameworks. However, as our diagnostic tests confirm, these models might be misspecified for this index; their inherent assumption of a linear process cannot accommodate the abrupt discontinuities and long-memory volatility clustering. Consequently, as evidenced by the AR benchmark's out-of-sample performance, these models produce forecasts that systematically fail to capture turning points and inflection moments triggered by regulatory shifts. The second stream has adopted machine learning algorithms, such as LSTM networks and support vector regression, which are better equipped to model non-linearity. Yet these approaches introduce a different set of inadequacies for forecasting of this particular series. LSTM models, despite their sequential learning capabilities, are sensitive to hyperparameter specification and training length selection, and their performance degrades markedly when confronted with the non-stationary volatility regimes characteristic of this index. Support vector regression, while flexible, employs a loss function that may not optimally balance the trade-off

between fitting the index's pronounced volatility spikes and its gradual trend evolutions. The benchmark results substantiate these limitations, with LSTM and SVR yielding out-of-sample RMSE values substantially higher than those achieved by our proposed framework. Regression trees demonstrate even greater difficulty in generalizing beyond their training regimes. Critically, none of these existing approaches incorporate a systematic, data-driven mechanism for selecting and combining covariance structures that can simultaneously capture the multiple temporal scales–from multi-week cyclicality to persistent policy-induced shifts–that characterize this index. Thus, a clear gap persists: the literature lacks a forecasting framework that is simultaneously capable of flexibly adapting to the index's non-linear and non-stationary dynamics without parametric misspecification, while also incorporating a principled, automated approach to hyperparameter calibration that ensures robust out-of-sample generalization across the distinct regulatory and economic regimes embedded in the dataset.

4 Methodology

In this study, we employ a sophisticated forecasting technique, Gaussian Process Regression (GPR), to generate precise point forecasts for future vegetable prices using their historical values. The core intuition behind GPR is that it is a highly flexible, pattern-recognition engine. Unlike a standard model that imposes a fixed mathematical shape (such as a straight line or a specific curve) on the data, GPR adapts its shape to the complex rhythms inherent in agricultural price series–seasonal cycles, trends, and irregular fluctuations. It operates on a principle of similarity: it assumes that prices at points in time that are close together (e.g., this week and next week) are likely to be more similar than prices far apart. The model's key component is a "similarity function" that it uses to weigh the influence of all past prices when making a new forecast. To ensure the model captures the correct patterns–distinguishing a seasonal swing from random noise, for example–we carefully tune its internal settings, known as hyper-parameters. This tuning is performed automatically using Bayesian optimization, a search process that efficiently tests many possible configurations to find the one that would have most accurately predicted past data, rigorously assessed through a cross-validation procedure that mimics sequential forecasting on historical sub-periods. The final result is a tailored model that learns the unique temporal signature of the commodity's price history and uses that learned signature to produce a single, best-estimate forecast for future time points. This approach provides the Vegetable Basket Project with a powerful, data-driven tool for anticipating price movements based on observed market history, supporting more informed and proactive market management.

Gaussian process regression (*GPR*) represents a flexible Bayesian nonparametric framework wherein prior distributions are directly imposed on latent functions through carefully designed covariance structures. Via intentional selection among various kernel families, researchers encode substantive presuppositions regarding functional continuity, periodic dynamics, and characteristic input-dependency scales. This inherent adaptability has yielded significant predictive improvements across numerous domains, including commodity price projection, materials science, thermal system optimization, and long-term climate simulation (Zhang and Xu 2020; Jin and Xu 2025gg; Bisht et al. 2022; Jin and Xu 2025hh; Zhang and Xu 2020; Jin and Xu 2025ii; Liu et al. 2020). Formally, given N data points $(x_i, y_i)_{i=1}^{N}$,

where $x_i \in \mathbb{R}^d$ (e.g., feature vectors containing ten historical lagged wholesale agricultural commodity composite price indices of the Vegetable Basket Project) and $y_i \in \mathbb{R}$ (the corresponding daily index value), *GPR* postulates the underlying mapping $f(\cdot)$ follows a Gaussian stochastic process prior. Consequently, any finite set of inputs x_i generates a joint multivariate normal distribution governing latent functional outputs $f(x_i)$ (Jin and Xu 2025jj; Neal 1997). Observational uncertainty is integrated by augmenting this prior with independent Gaussian noise components, producing a unified probabilistic architecture encompassing latent function evaluations and observed responses. *GPR* effectiveness critically relies upon covariance kernel design: Foundational kernels include squared-exponential (radial basis), Matern, rational-quadratic, and periodic forms; richer behavioral patterns emerge via additive or multiplicative kernel combinations, facilitating representation of phenomena like nonstationary volatility and structural discontinuities prevalent within financial temporal sequences (Jin and Xu 2025kk; Zhang and Xu 2020; Jin and Xu 2025ll; Tu et al. 2024; Jin and Xu 2025mm). Enhanced expressive power may be introduced via deterministic basis expansions $b(x)$–e.g., polynomial transformations or random Fourier features–projecting inputs into enriched Hilbert spaces to model complex relationships (Zhang and Xu 2020; Jin and Xu 2025nn; Kou et al. 2015; Jin and Xu 2025oo). Hyperparameter estimation–including kernel length-scales (controlling smoothness), signal-variance parameters (governing output magnitude), and noise-variance terms (quantifying measurement imprecision)–typically advances through maximization of the log-marginal likelihood, intrinsically balancing data conformity against model parsimony. Alternatively, fully Bayesian implementations impose hierarchical hyperpriors, propagating uncertainty throughout posterior distributions. A primary benefit of *GPR* resides in analytically tractable expressions for posterior predictive moments, enabling direct derivation of point forecasts and credible intervals without resorting to sampling algorithms (Williams and Rasmussen 1995; Neal 1997). Such rigorous uncertainty quantification proves especially valuable for informing robust agricultural pricing policy decisions. However, cubic time complexity $\mathcal{O}(N^3)$ for matrix inversion and $\mathcal{O}(N^2)$ memory demands present computational constraints for extensive or high-frequency datasets. Mitigation approaches incorporate sparse approximations, inducing-point techniques, or stochastic variational inference, which develop efficient surrogate models using representative subsets or variational distributions, thus extending *GPR*'s utility to large-scale or streaming data contexts. By combining kernel-based interpretability with comprehensive Bayesian uncertainty propagation, *GPR* provides a potent methodology for modeling nonlinear, nonstationary, and volatile dynamics inherent to wholesale agricultural commodity composite price evolution of the Vegetable Basket Project, delivering both accurate predictions and statistically calibrated uncertainty metrics indispensable for evidence-based decision-making by market participants and regulatory institutions.

Gaussian process (*GP*) modeling establishes the probabilistic underpinning of this study by assigning Bayesian priors directly to infinite-dimensional functional spaces. Rather than predefining an output distribution, one specifies a mean function $m(x) = E[l(x)]$ and a covariance kernel $k(x, x') = \mathrm{Cov}[l(x), l(x')]$, which collectively determine the latent stochastic process $f(x)$ (Zhang and Xu 2020; Jin and Xu 2025pp; Crosby et al. 2016; Jin and Xu 2025qq). Under the standard zero-mean *GP* postulate, any finite input set x_i generates a multivariate normal distribution whose covariance matrix entries originate from $k(x_i, x_j)$. In regression contexts, observations conform to $y = b(x)^T\beta + l(x)$, where $b(x) \in \mathbb{R}^p$ denotes predefined basis functions–such as polynomial expansions or spline representations–and β signifies the corresponding coefficient vector. For projecting composite wholesale agri-

cultural commodity price indicators of the Vegetable Basket Project, $b(x)$ incorporates ten preceding daily values to capture short-term autocorrelation and temporal persistence. All model parameters–encompassing basis coefficients β, noise variance σ^2, and kernel hyperparameters θ controlling $k(\cdot,\cdot \mid \theta)$–experience simultaneous estimation through marginal likelihood optimization, ensuring cohesive, data-driven calibration of uncertainty and trend components. Our investigation assesses two principal kernel formulations: The first utilizes an isotropic kernel regulated by a single length-scale parameter l, imposing uniform smoothness across input dimensions. Conversely, automatic relevance determination (ARD) frameworks allocate distinct length-scales l_m to individual covariates, enabling differential predictor weighting according to explanatory significance. To guarantee positivity, scale parameters undergo logarithmic transformation–e.g., $\theta_1 = \log \sigma_l, \theta_2 = \log \sigma_f$ for isotropic configurations (Zhang and Xu 2021; Jin and Xu 2025rr; Zeng et al. 2020; Jin and Xu 2025ss). For ARD implementations, this extends to $\theta = (\log \sigma_1, \log \sigma_2, ..., \log \sigma_d, \log \sigma_f)$, maintaining strictly positive length-scales and signal variances (Zhang and Xu 2021; Jin and Xu 2025tt; Yadav et al. 2024; Jin and Xu 2025uu). When combined with the fixed basis expansion $b(x)$, these covariance frameworks model intricate nonlinear interdependencies among lagged inputs–critical for representing sudden policy-induced discontinuities, seasonal demand variations, and structural regime shifts in composite agricultural indices (Zhang and Xu 2021; Jin and Xu 2025vv; Richardson et al. 2017; Jin and Xu 2025). The Bayesian *GP* paradigm yields closed-form expressions for posterior moments of latent functions conditional on empirical data, permitting direct derivation of point predictions and credible intervals absent sampling-based approximations like Markov chain Monte Carlo (Williams and Rasmussen 1995; Neal 1997). This analytical tractability provides stakeholders with accurate forecasts accompanied by statistically rigorous uncertainty quantification–particularly crucial in agricultural price analytics where regulatory actions or meteorological disturbances may provoke heavy-tailed deviations and structural breaks (Zhang and Xu 2021; Yeardley et al. 2020; Zhang and Xu 2020; Yoo and Jung 2024). By integrating flexible covariance architectures, likelihood-driven hyperparameter optimization, and coherent Bayesian inference, the proposed *GP* methodology constitutes a versatile analytical instrument capable of processing high-dimensional inputs, accommodating extreme-value phenomena, and dynamically adapting to evolving market structures. Such adaptability positions *GP* models as resilient analytical resources for regulators and market participants pursuing evidence-based, uncertainty-aware insights into composite wholesale agricultural commodity price dynamics, featuring inherent capacity for iterative enhancement amid changing economic landscapes.

$$\text{Isotropic Exponential: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 e^{-\frac{r}{\sigma_l}} \tag{1}$$

$$\text{Isotropic Squared Exponential: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 e^{-\frac{1}{2}\frac{(x_i-x_j)^T(x_i-x_j)}{\sigma_l^2}} \tag{2}$$

$$\text{Isotropic Matern } 5/2\text{: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 \left(1 + \frac{\sqrt{5}r}{\sigma_l} + \frac{5r^2}{3\sigma_l^2}\right) e^{-\frac{\sqrt{5}r}{\sigma_l}} \tag{3}$$

$$\text{Isotropic Rational Quadratic: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 \left(1 + \frac{r^2}{2\alpha\sigma_l^2}\right)^{-\alpha} \tag{4}$$

$$\text{Isotropic Matern } 3/2\text{: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 \left(1 + \frac{\sqrt{3}r}{\sigma_l}\right) e^{-\frac{\sqrt{3}r}{\sigma_l}} \tag{5}$$

$$\text{Nonisotropic Exponential: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 e^{-\sqrt{\sum_{m=1}^{d} \frac{\left(x_{im}-x_{jm}\right)^2}{\sigma_m^2}}} \tag{6}$$

$$\text{Nonisotropic Squared Exponential: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 e^{-\frac{1}{2}\sum_{m=1}^{d} \frac{\left(x_{im}-x_{jm}\right)^2}{\sigma_m^2}} \tag{7}$$

$$\text{Nonisotropic Matern } 5/2\text{: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 \left(1 + \sqrt{5\sum_{m=1}^{d} \frac{\left(x_{im}-x_{jm}\right)^2}{\sigma_m^2}} + \frac{5}{3}\sum_{m=1}^{d} \frac{\left(x_{im}-x_{jm}\right)^2}{\sigma_m^2}\right) e^{-\sqrt{5\sum_{m=1}^{d} \frac{\left(x_{im}-x_{jm}\right)^2}{\sigma_m^2}}} \tag{8}$$

$$\text{Nonisotropic Rational Quadratic: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 \left(1 + \frac{1}{2\alpha}\sum_{m=1}^{d} \frac{\left(x_{im}-x_{jm}\right)^2}{\sigma_m^2}\right)^{-\alpha} \tag{9}$$

$$\text{Nonisotropic Matern } 3/2\text{: } k\left(x_i, x_j|\theta\right) = \sigma_f^2 \left(1 + \sqrt{3\sum_{m=1}^{d} \frac{\left(x_{im}-x_{jm}\right)^2}{\sigma_m^2}}\right) e^{-\sqrt{3\sum_{m=1}^{d} \frac{\left(x_{im}-x_{jm}\right)^2}{\sigma_m^2}}} \tag{10}$$

A methodical comparative evaluation is performed for four unique basis-function configurations–formally specified in Eq. (11)–(14)–incorporated within our forecasting framework. To guarantee unbiased assessment, initial temporal predictors experience transformation into three distinct matrix formulations: (i) the primary predictor matrix $X = (x_1,\ x_2,\ \ldots\ , x_n)'$; (ii) its squared-element counterpart $X^2 = \begin{pmatrix} x_{11}^2 & x_{12}^2 & \cdots & x_{1d}^2 \\ x_{21}^2 & x_{22}^2 & \cdots & x_{2d}^2 \\ \vdots & \vdots & \vdots & \vdots \\ x_{T1}^2 & x_{T2}^2 & \cdots & x_{Td}^2 \end{pmatrix}$; and (iii) the basis-augmented design matrix $B = \left(b\left(x_1\right),\ b\left(x_2\right),\ \ldots\ , b\left(x_n\right)\right)'$, where every row $b(x_i)$ represents the transformed feature vector. Analysis of these constructs facilitates empirical quantification of each expansion's influence on representational adequacy, computational burden, and parametric economy. The foundational approach–integrating exhaustive pairwise quadratic interactions–systematically incorporates products among all possible pairs of the d predictor variables. While capturing localized second-order dependencies, it considerably enlarges the feature space, increasing memory requirements and optimization duration. The secondary formulation–a polynomial basis of elevated degree–includes monomials beyond quadratic order; despite efficacy in characterizing intricate nonlinearities, it incurs significant overparameterization hazards, necessitating stringent regularization (e.g., ℓ_1 or ℓ_2 constraints) for dependable generalization. The third methodology–a judiciously constrained linear-quadratic hybrid–merges first-order components with a curated subset of interactions identified through domain knowledge or feature selection algorithms. This moderated strategy maintains intermediate nonlinear flexibility while decreasing parameter dimensionality and alleviating multicollinearity issues. Ultimately, orthonormal decomposition techniques–encompassing principal component analysis and related spectral meth-

ods–project the original d-dimensional inputs onto a compressed ensemble of uncorrelated latent factors, thereby enhancing numerical conditioning, reducing collinear dependencies, and achieving dimensionality reduction. Empirical validation employs high-dimensional, nonstationary pricing series–a domain demanding strong interpretability–yielding specific guidance concerning basis selection and hyperparameter calibration. The resultant modular framework preserves theoretical consistency while adapting to volatile financial contexts. Consequently, it furnishes analysts and policymakers with explicit insights regarding trade-offs between computational efficiency, parametric economy, and interpretative transparency–especially critical during forecasting amidst structural transitions or regulatory evolution–while readily assimilating specialized knowledge across diverse market regimes.

$$\text{Empty: } B = Empty\ \ Matrix \tag{11}$$

$$\text{Constant: } B = I_{n\times 1} \tag{12}$$

$$\text{Linear: } B = [1, X] \tag{13}$$

$$\text{Pure Quadratic: } B = \left[1, X, X^2\right] \tag{14}$$

Hyperparameter calibration advances via a stratified decuple cross-validation protocol coupled with Bayesian optimization employing the Expected Improvement Per Second Plus (*EIPSP*) acquisition criterion (Zhang and Xu 2021; Sheng et al. 2017; Zhang and Xu 2021; Liu et al. 2020). A Gaussian process (*GP*) serves as the probabilistic surrogate, approximating the functional relationship $f : \mathrm{x} \mapsto y$ by associating hyperparameter configurations x_i with performance outcomes y_i. Preceding optimization commencement, an initial space-filling design selects $N_s = 4$ candidate configurations within prescribed boundaries, ensuring comprehensive exploration of the hyperparameter domain (Zhang and Xu 2022; Kim et al. 2022; Jin and Xu 2024n). Each sampled configuration experiences automated diagnostic validation to evaluate *GP* numerical stability, assess convergence characteristics, and verify metric computation precision prior to Bayesian updating (Zhang and Xu 2021; Yang et al. 2018; Jin and Xu 2024o). During sequential iteration, the surrogate model undergoes re-estimation utilizing all assessed configurations, while novel proposals originate from *EIPSP* maximization. At iteration t, the posterior distribution $Q\left(f \mid \mathrm{x}_i, y_i \text{ for } i = 1, \ldots, T\right)$ updates incrementally, with subsequent hyperparameter vectors x_{t+1} determined by optimizing $EIPSP(\mathrm{x}) = EI(\mathrm{x}, Q)/\mu_S(\mathrm{x})$. Here, $EI(\mathrm{x}, Q) = \mathbb{E}_Q\left[\max\left(0, \mu_Q\left(\mathrm{x}_{\text{best}}\right) - f(\mathrm{x})\right)\right]$ represents standard Expected Improvement, $\mu_Q(\mathrm{x}_{\text{best}})$ indicates the current posterior mean optimum, and $\mu_S(\mathrm{x})$ denotes an auxiliary *GP* model forecasting computational duration using historical runtime records (Zhang and Xu 2021; Ma et al. 2022; Zhang and Xu 2022; Hu and Wang 2015)). By explicitly reconciling anticipated performance enhancements against projected computational expenses, *EIPSP* prioritizes configurations offering significant improvement with minimal resource expenditure. To prevent premature convergence, the algorithm tracks predictive uncertainty $\sigma_F(\mathrm{x})$. Should $\sigma_F(\mathrm{x})$ decrease below a threshold proportional to estimated noise variance σ_{PN} (scaled by $t_{\sigma_{PN}}$), all kernel length-scales undergo doubling ($\boldsymbol{\theta} \leftarrow 2\boldsymbol{\theta}$), broadening the surrogate's exploration capacity (Bull 2011; Zhang and Xu 2021; Jin and Xu 2024p)). Additionally, overly dense spatial clustering of successive candidates–suggesting stagnation–activates exponential length-scale inflation (capped at fivefold) to widen search radii and rebalance exploration-exploitation tradeoffs (Xu and Zhang 2022; Wang et al. 2018; Zhang and Xu 2021; Egrioglu et al. 2024)). Inte-

grating adaptive exploration-exploitation scheduling, dynamic kernel modification, and resource-sensitive acquisition within a spatiotemporal *GP* architecture, the *EIPSP* approach efficiently identifies optimal hyperparameters across irregular, high-dimensional response surfaces characteristic of wholesale agricultural commodity composite price index forecasting of the Vegetable Basket Project. This methodology accelerates convergence by favoring high-information, low-cost evaluations; counters overfitting by evading local optimum entrapment; and consequently improves generalization to unseen data environments. The resulting calibration infrastructure furnishes a scalable, theoretically sound toolkit for traversing complex, multimodal performance landscapes, enhancing both predictive dependability and computational economy in econometric applications involving nonstationary, high-frequency data streams. Its resilience against noisy, discontinuous objective functions proves particularly beneficial for modeling volatile commodity price series subject to regulatory shifts and supply-chain disruptions.

We present an integrated Bayesian optimization framework designed to methodically calibrate all critical elements of a Gaussian process regression forecasting workflow–including basis-function determination, covariance kernel hyperparameter estimation, observational noise variance quantification, and optimal data-normalization strategy selection–explicitly targeting enhanced generalization capability (Zhang and Xu 2022; Chandiwana et al. 2021). Fundamental to this methodology resides a probabilistic surrogate, typically a Gaussian process, which infers the functional relationship between candidate hyperparameter vectors and associated predictive performance outcomes. Optimization progresses iteratively: following assessment of candidate configurations via cross-validation, the surrogate incorporates new performance evidence, and subsequent proposals derive from maximizing a designated acquisition function (e.g., expected improvement). This cyclical process intrinsically balances the exploration-exploitation trade-off–investigating underexplored hyperparameter regions while capitalizing on areas showing promising results–thereby directing convergence toward configurations demonstrating superior out-of-sample performance. To comprehensively evaluate forecasting quality, we utilize four complementary performance indicators: (1) Relative Root-Mean-Square Error (*RRMSE*): $RRMSE = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i^{obs}-y_i^{for})^2}}{\frac{1}{n}\sum_{i=1}^{n} y_i^{obs}}$, scaling standard *RMSE* by mean observed values to enable scale-invariant comparisons across heterogeneous series (Jamieson et al. 1991; Heinemann et al. 2012; Li et al. 2013; Despotovic et al. 2016); (2) Mean Absolute Error (*MAE*): $MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i^{obs} - y_i^{for}|$, quantifying average absolute forecast deviation while mitigating disproportionate outlier influence; (3) Root-Mean-Square Error (*RMSE*): $RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i^{obs} - y_i^{for})^2}$, applying quadratic penalization to accentuate larger forecasting inaccuracies; (4) Correlation Coefficient (*CC*): $CC = \frac{\sum_{i=1}^{n}\left(y_i^{\mathrm{obs}}-\overline{y^{\mathrm{obs}}}\right)\left(y_i^{\mathrm{for}}-\overline{y^{\mathrm{for}}}\right)}{\sqrt{\sum_{i=1}^{n}\left(y_i^{\mathrm{obs}}-\overline{y^{\mathrm{obs}}}\right)^2\sum_{i=1}^{n}\left(y_i^{\mathrm{for}}-\overline{y^{\mathrm{for}}}\right)^2}}$, assessing directional concordance and magnitude correspondence between predicted and empirical values. Here, n denotes observation count, y^{obs} and y^{for} signify observed and forecasted values respectively, while overbars indicate arithmetic means. Simultaneous optimization across *RRMSE*, *MAE*, *RMSE*, and *CC* produces a balanced performance profile–ensuring scale invariance, attenuating outlier sensitivity, and preserving temporal coherence. This multi-metric integration proves indispensable when modeling complex, high-dimensional nonstationary sequences like wholesale agricultural commodity composite price indices of the Vegetable Basket Project, where single-metric dependence might mask latent model shortcomings.

Consequently, this framework provides rigorous extrapolative validation and establishes a transparent, theoretically grounded hyperparameter selection mechanism, yielding forecasts robust to intricate, evolving commodity price dynamics. Explicit quantification of trade-offs between distinct accuracy dimensions further augments interpretability for policy-relevant applications necessitating reliable uncertainty characterization in decision-critical contexts. Analysis here is implemented via a desktop computer with Intel(R) Core(TM) i7-9700 CPU and 32GB RAM. The runtime for model training and out-of-sample forecasting through the MATLAB platform is about 3 hours.

5 Results

This research implements a meticulously structured, dual-phase experimental design to evaluate the predictive efficacy of a Gaussian Process Regression (*GPR*) forecasting framework for wholesale agricultural commodity composite price indices of the Vegetable Basket Project. The preliminary in-sample calibration stage encompasses training the model using daily pricing observations spanning September 27, 2005, to February 16, 2022. Hyperparameter tuning is conducted autonomously within each segment of a stratified, decile-based cross-validation methodology. This strategy partitions observations into ten non-overlapping subsets, deliberately aligned with quantiles of the response variable distribution. Such intentional stratification guarantees every validation fold incorporates a representative spectrum of distinct regulatory environments, structural market transitions, and diverse volatility regimes (Zhang and Xu 2021; Sheng et al. 2017). Quantile-based segmentation effectively addresses inherent temporal heterogeneity within the pricing sequence, simultaneously reducing susceptibility to overfitting during anomalous or atypical temporal intervals. Following identification of optimal hyperparameters across all decile partitions, the out-of-sample prediction phase initiates. An ensemble technique is deployed, integrating ten independently configured *GPR* models. Each component model undergoes training employing nine folds, with predictive validation executed on the excluded tenth fold. This ensemble produces sequential one-step-ahead projections across an extensive evaluation horizon covering 1,146 days (February 17, 2022, to April 10, 2025). As illustrated in Fig. 2, the optimization trajectory directed by the Expected Improvement Per Second Plus (*EIPSP*) acquisition function consistently converged toward three identical design specifications universally among all partitions: (i) implementation of non-normalized feature matrices; (ii) adoption of an isotropic Matern 5/2 covariance kernel (Equation 3); and (iii) utilization of a constant basis function (Equation 12). This configuration uniformity strongly signifies the collective efficacy of these elements in capturing fundamental dynamics underlying the aggregate agricultural commodity pricing indicator of the Vegetable Basket Project. Comprehensive hyperparameter estimates–incorporating length scales, signal variances, and noise variances for validation segments CV1 through CV10–are methodically documented in Table 2. Segmenting the chronological data sequence into deciles fulfills two primary objectives. Primarily, it verifies that selected kernel and mean function configurations maintain reliable out-of-sample forecasting precision despite encountering policy revisions, methodological refinements, and transient market disturbances, demonstrating the *GPR* architecture's resilience to institutional transformations. Secondarily, exposing the probabilistic surrogate to varied structural discontinuities during optimization augments its adaptability to underlying

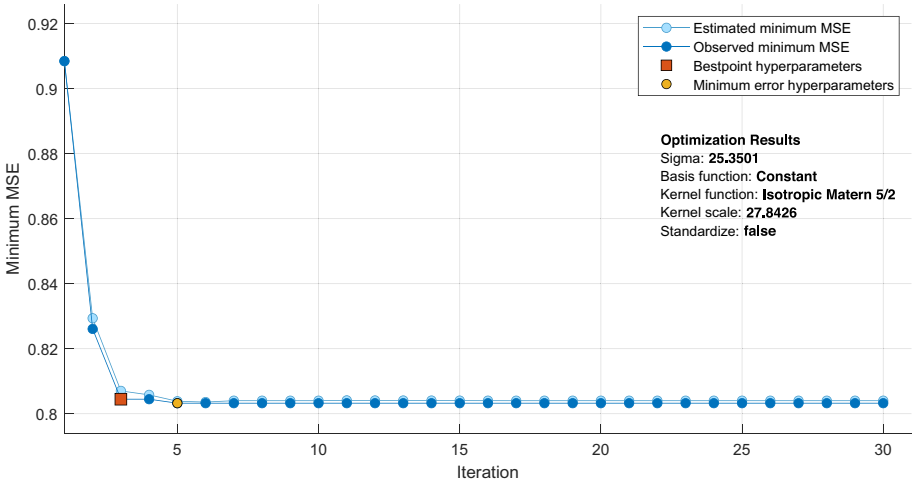


Fig. 2 Results of Bayesian optimisations

regime shifts, consequently fortifying extrapolative stability. Through rigorous examination across heterogeneous price pathways within each partition, the resultant *GPR* ensemble achieves superior forecasting fidelity. It attains this enhanced performance while preserving computational tractability, even when processing high-dimensional and nonstationary attributes characteristic of agricultural market datasets. Collectively, this dual-phase validation infrastructure provides a stringent assessment of the methodology's reliability, adaptability, and practical utility for projecting trajectories of composite agricultural commodity price indices of the Vegetable Basket Project within dynamically evolving economic landscapes. The stratified partitioning protocol ensures robustness against unforeseen structural disruptions, while the ensemble methodology generates probabilistically quantified projections. These forecasts exhibit essential properties for informing policy-sensitive decision processes within inherently volatile commodity markets.

The predictive performance of the Gaussian Process Regression (*GPR*) ensemble received comprehensive assessment across an uninterrupted 1,146-day out-of-sample forecasting horizon, spanning February 17, 2022, to April 10, 2025. Ten distinct *GPR* instantiations–each independently optimized through stratified decile-based cross-validation (designated CV1–CV10 in Table 2)–generated sequential one-step-ahead projections for composite wholesale agricultural commodity price indicators daily of the Vegetable Basket Project. Utilizing multiple surrogate models counters particular partialities inherent to individual configurations while maintaining chronological integrity of foundational data. For every day within the testing period, the ten separate predictions underwent consolidation via arithmetic averaging, producing a unified ensemble forecast (Timmermann 2006; Costantini et al. 2017). The ensemble's proficiency in accurately tracking both gradual trend evolutions and abrupt structural discontinuities receives visual confirmation in Fig. 3, while Fig. 4 illustrates corresponding percentage forecasting deviations. Quantification of predictive accuracy employed four established statistical metrics: Relative Root-Mean-Square Error (*RRMSE*), Root-Mean-Square Error (*RMSE*), Mean Absolute Error (*MAE*), and Correlation Coefficient (*CC*). Documented outcomes within Table 3 yielded $RRMSE = 0.2040\%$,

Table 2 Results of model parameter estimations

	CV1	CV2	CV3	CV4	CV5	CV6	CV7	CV8	CV9	CV10
σ	0.907697710	0.927784404	0.842885817	0.899788531	0.911562592	0.886066255	0.847718955	0.813826980	0.943478544	0.881873345
β	186.752955237	101.164896890	272.997951727	142.302643044	180.611881418	−1.281775078	28.805200302	55.925458059	64.093836666	158.351938116
σ_l	1330.917061043	2669.480703726	1263.826108574	1209.199850070	2478.989190266	1316.411712398	1631.727619155	3441.989483897	4922.170564102	1236.171195038
σ_f	360.971831931	742.197785494	353.598839799	274.221617275	625.879842298	349.305377512	470.999659514	897.792230820	1338.216901515	327.978469517

$RMSE = 0.4841$, $MAE = 0.3430$, and $CC = 99.929\%$. The foundational decile-aligned cross-validation technique systematically partitions the chronological dataset into ten non-overlapping training-validation pairs, deliberately incorporating diverse volatility regimes and distinct regulatory contexts. This architectural design guarantees every surrogate model encounters heterogeneous market behavior, empirically validating robustness of selected covariance kernels and mean function specifications when confronting policy revisions, methodological innovations, and transient market disruptions. Requiring each component model to generalize from its specific calibration subset substantially reduces vulnerability to localized overfitting. Furthermore, synthesizing predictions via ensemble averaging during operational forecasting reduces errors attributable to any single constituent and diminishes anomalous projection impacts, consequently enhancing resilience against unforeseen macroeconomic disturbances or sudden regulatory changes. Collectively, integrating probabilistic surrogate modeling, systematic hyperparameter calibration via cross-validation, and predictive aggregation furnishes compelling evidence that *GPR*-based ensembles provide consistently dependable projections within inherently nonstationary, high-dimensional composite wholesale agricultural commodity pricing environments of the Vegetable Basket Project. This synthesis of sophisticated machine learning methodologies with stringent validation protocols yields interpretable, adaptable, and statistically rigorous predictions. The framework's ability to quantify forecast uncertainty while preserving temporal coherence offers distinct utility for central banking authorities and regulatory bodies requiring actionable intelligence amid volatile market conditions, thereby supporting evidence-based monetary policy determinations and supply-chain risk mitigation strategies.

This study implements a comprehensive suite of statistical diagnostics on one-step-ahead forecast residuals, rigorously validating internal coherence of the Gaussian Process Regression (*GPR*) forecasting methodology. For every residual sequence, standardized autocorrelation functions were computed across lags 1–20, adhering to established econometric protocols ensuring methodological transparency and reproducibility within temporal sequence analysis (Ou and Wang 2011a, b, c; Han and Zhang 2015). As demonstrated in Fig. 5, no computed autocorrelation coefficient attained statistical significance, indicating forecast errors exhibit properties consistent with white noise stochastic processes. This absence of detectable serial dependence implies the selected mean function specification adequately captures dominant systematic variations inherent to wholesale agricultural commodity composite price index sequences of the Vegetable Basket Project. Nevertheless, undetectable autocorrelation does not preclude potential heteroskedastic patterns evolving temporally. To address possible volatility clustering within residuals, subsequent *GPR* refinements could incorporate explicit conditional variance parameterization. Illustratively, embedding Autoregressive Conditional Heteroskedasticity (ARCH) frameworks–or generalized variants like Glosten-Jagannathan-Runkle (GJR-GARCH) or Bollerslev's Generalized ARCH (GARCH)–at residual levels would enable direct estimation of time-varying error dispersion. More sophisticated approaches, such as Exponential GARCH (EGARCH) or Stochastic Volatility (SV) models, would characterize leverage effects and persistent volatility clustering, capturing asymmetric market reactions to positive versus negative informational innovations. An EGARCH parameterization explicitly quantifies differential shock impacts on volatility dynamics through its logarithmic conditional variance specification, while SV methodologies introduce latent stochastic processes governing variance evolution. Integrating such advanced heteroskedasticity treatments within *GPR*'s inher-

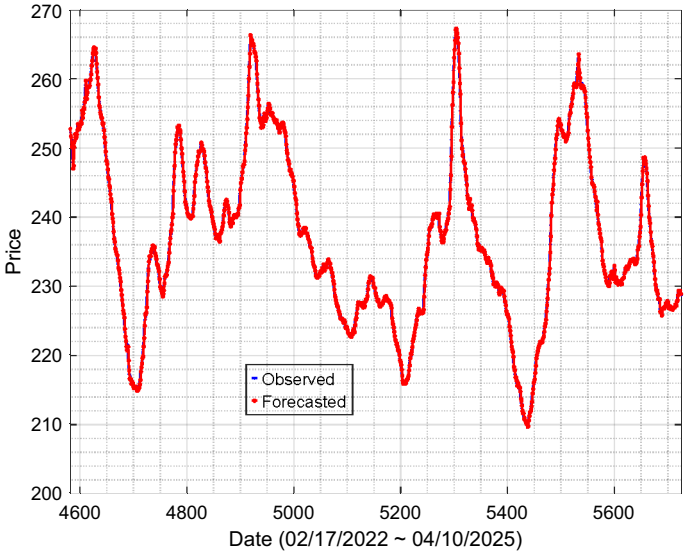


Fig. 3 Comparisons between predicted and actual composite wholesale agricultural commodity price indices of the Vegetable Basket Project during the 02/17/2022–04/10/2025 out-of-sample period

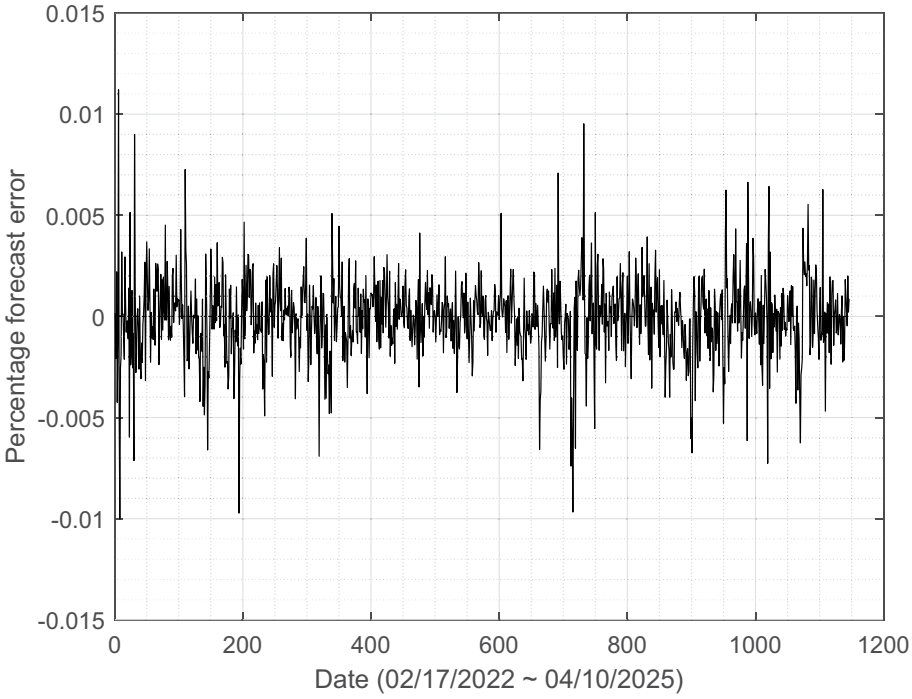


Fig. 4 Percentage prediction errors of composite wholesale agricultural commodity price indices of the Vegetable Basket Project during the 02/17/2022–04/10/2025 out-of-sample period

Table 3 Prediction performance

Period	*RRMSE*	*RMSE*	*MAE*	*CC*
In-sample training phase (09/27/2005–02/16/2022)	0.4930%	0.9601	0.6288	99.954%
Out-of-sample testing phase (02/17/2022–04/10/2025)	0.2040%	0.4841	0.3430	99.929%

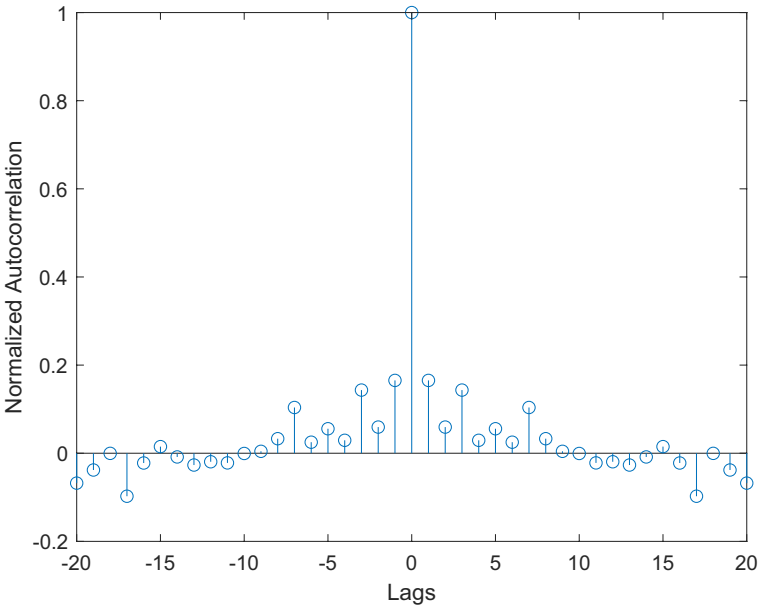


Fig. 5 Results of error autocorrelation analysis

ent nonparametric flexibility could generate more nuanced representations of regulatory discontinuities and speculative market episodes. This integration would yield adaptively scaled prediction intervals responsive to shifting market turbulence intensities. Such methodological synthesis potentially delivers forecasts that are expectationally unbiased while accurately reflecting intrinsic risk structures of composite agricultural commodity price trajectories. Consequently, these enhanced predictions would possibly provide greater utility for policy formulation requiring rigorous forecast uncertainty quantification.

6 Sensitivity analysis

We conducted another experiment utilizing a reduced lag structure of three temporal inputs. The empirical results were decisive: this parsimonious model configuration resulted in an approximate 30% increase in the RMSE during the out-of-sample testing period compared to the model with ten lags. This degradation in forecasting performance indicates that while a three-lag model might help mitigate potential multicollinearity issues, it could possibly simultaneously incur an underfitting cost. The superior performance of the ten-lag specification suggests that the price discovery process for the composite Vegetable Basket index

Table 4 Prediction performance: Robustness check with the rolling window validation approach

Period	*RRMSE*	*RMSE*	*MAE*	*CC*
In-sample training phase (09/27/2005–02/16/2022)	0.4928%	0.9598	0.6286	99.954%
Out-of-sample testing phase (02/17/2022–04/10/2025)	0.2042%	0.4845	0.3431	99.928%

might be influenced by a more extended temporal dependency, capturing complex, longer-memory dynamics such as multi-week cyclicality and the persistent effects of supply chain adjustments. We acknowledge that the ten-lag model possesses a higher parametric capacity; however, the rigorous Bayesian optimization process employed in our study was explicitly designed to regularize the model and mitigate overfitting. The fact that this more complex model demonstrated robust generalization and superior accuracy on unseen data provides strong empirical justification for its selection, indicating that the captured dependencies are genuine characteristics of the data-generating process.

We also conducted robustness check by considering the rolling window validation approach during the model training phase of 09/27/2005–02/16/2022. The resultant forecasting performance for both the in-sample training phase (09/27/2005–02/16/2022) and out-of-sample testing phase (02/17/2022–04/10/2025) is reported in Table 4. As compared to the results in Table 3, we can see that the performance metrics are generally robust.

7 Benchmark analysis

A detailed benchmarking analysis was performed to evaluate the Gaussian process regression (GPR) framework against a selection of four alternative modeling methodologies. This comparative assessment incorporated several widely-used machine learning algorithms, specifically the long short-term memory (LSTM) neural network, support vector regression (SVR), and regression trees (RT). Furthermore, a traditional econometric time-series approach, the autoregressive (AR) model, commonly applied in economic prediction contexts, was also included in the evaluation. To ensure an equitable and rigorously controlled comparison of the core modeling architectures, the experimental setup was carefully designed. The competing LSTM, SVR, RT, and AR models were intentionally parameterized to utilize the exact same set of input variables as the GPR framework. This methodological consistency is critical, as it ensures that any observed disparities in model performance can be directly attributed to the unique algorithmic capabilities of each technique rather than to differences in input data. During the subsequent out-of-sample testing phase, the calculated root mean square error (RMSE) metrics for the LSTM, SVR, RT, and AR models were 0.7859, 0.9629, 1.2519, and 1.4093, respectively. It is noteworthy that all these benchmark model RMSE values are higher than the 0.4841 error rate achieved by the GPR technique. This significant performance gap conclusively illustrates that the GPR approach provides markedly superior predictive accuracy for the specific composite wholesale price indices under investigation. The pronounced benefit of GPR implies that its inherent capacity for effectively capturing complex, nonlinear temporal dependencies is particularly well-suited to this forecasting challenge. Consequently, it demonstrates a clear advantage, outstripping the capabilities of not only simpler linear statistical models but also other, more sophisticated machine learning methods included in the study.

The quantitative underperformance of the benchmark models, as evidenced by their higher out-of-sample RMSE values relative to the GPR ensemble, is not merely a matter of degree but stems from fundamental methodological inadequacies when applied to the specific characteristics of the Vegetable Basket Project index. The autoregressive (AR) model, as a representative of the classical econometric time-series family that includes ARIMA and VAR, is predicated on assumptions of linearity. However, as demonstrated by the Brock-Dechert-Scheinkman (BDS) test results, the index exhibits significant non-linear dependencies that such linear frameworks cannot capture. Furthermore, the index's journey from 2005 to 2025 has been punctuated by breaks–episodes of regulatory change, sectoral transformation, and macroeconomic upheaval–that render any assumption of a stable, linear data-generating process invalid. While Vector Autoregressions (VAR) can model multivariate interactions, they remain linear and, when applied to a univariate forecasting task with lagged terms, they share the same core limitation of being unable to adapt their functional form to abrupt shifts in regime or non-linear volatility clustering. The machine learning benchmarks, while more flexible, present a different set of inadequacies. Support Vector Regression (SVR) can model non-linearity, but it is fundamentally a discriminative model that provides point forecasts without potential for a principled, probabilistic measure of uncertainty. The Long Short-Term Memory (LSTM) network, while powerful for sequence learning, operates as a "black box" with numerous hyperparameters that are challenging to tune for optimal generalization on a series with non-stationary characteristics. Its performance is highly sensitive to the length of the training sequence and the risk of overfitting to the most recent, but not necessarily most representative, volatility regime. Moreover, LSTM, like SVR and Regression Trees (RT), does not inherently provide the well-calibrated, analytically tractable predictive performance that is a hallmark of the Bayesian GPR framework. The GPR model, by contrast, is explicitly designed to address the challenge: its Bayesian non-parametric nature allows it to adapt to non-linearities and structural shifts through its flexible kernel structure. Thus, the empirical performance gap is a direct reflection of these deeper methodological alignments–or misalignments–with the complex and non-stationary nature of the Vegetable Basket Project price index.

8 Policy implications

The empirical validation of our Gaussian Process Regression (GPR) ensemble, achieving a high correlation (CC = 0.99929) and minimal error (RRMSE = 0.2040%) in forecasting the composite wholesale price index of the Vegetable Basket Project, is not merely a technical accomplishment. It represents a foundational advancement with profound, actionable implications for the entire ecosystem of agricultural policy in China. The transition from declarative potential to analytical utility hinges on the specific capabilities of our model: its probabilistic output, its adaptability to structural breaks, and its precise tracking of both trends and volatilities. This section delineates how these capabilities can be systematically integrated into the decision-making frameworks of various stakeholders to enhance market stability, food security, and sustainable resource allocation. For production planning and crop portfolio management, the GPR model's forecasts provide a dynamic, data-driven signal that transcends traditional, backward-looking price analyses. The model's ability to generate not just point forecasts but full predictive distributions if needed allows agricultural

extension services and individual farmers to assess both the most likely future price scenario and the associated risks. For instance, a forecast indicating a high probability of a sustained price increase for the composite index over the next planting season, coupled with narrow credible intervals, would signal a strong incentive for farmers to maintain or slightly expand production of constituent commodities. More critically, the model can signal impending downturns. A forecast projecting a high probability of a significant price decline, perhaps triggered by early indicators of a supply glut or weakening demand, would enable preemptive action. Policymakers could use this lead time to activate programs that incentivize crop diversification or temporary acreage reduction for specific, over-supplied commodities within the basket, thereby mitigating the revenue shock for farmers and preventing inefficient allocation of resources like water and fertilizer. This moves policy from a reactive, post-crisis support model to a proactive, risk-based planning paradigm. In the realm of supply chain logistics and inventory control, the implications are equally transformative. The demonstrated low Mean Absolute Error (MAE = 0.3430) signifies the model's reliability in predicting daily price levels, which is crucial for operational decisions. Logistics operators and wholesale distributors can integrate these forecasts into their inventory management systems to optimize storage and transportation. A predicted upward price trajectory would justify the costs of strategic inventory accumulation, ensuring supply continuity for processors and retailers. Conversely, a forecasted price decline would advise against holding large inventories, reducing carrying costs and spoilage risks, particularly for perishable fruits and vegetables within the basket. Furthermore, the model's capacity to capture abrupt structural discontinuities is vital for managing the fallout from supply chain disruptions. An early warning of a price spike following a climatic event or a logistics bottleneck allows for the strategic release of goods from distributed warehouses to specific markets, smoothing out inter-regional price disparities and curbing inflationary pressures before they cascade to consumers. This enhances the overall resilience of the food supply system against both endogenous market dynamics and exogenous shocks. The management of strategic national reserves is a cornerstone of China's food security policy, and our GPR framework offers a quantitative basis for optimizing these interventions. Currently, decisions to release or stockpile commodities are often based on contemporaneous price thresholds. Our model introduces a forward-looking dimension. By running conditional simulations under different intervention scenarios if pursued, policymakers can conduct counterfactual analyses. For example, they could model the price effect of releasing a certain quantity of reserves at a future date, based on the forecasted price path. This allows for the calibration of both the timing and the scale of intervention to maximize market-stabilizing impact while conserving finite reserve stocks. The probabilistic nature of the forecast is key here; if the credible interval around a forecasted price spike is wide, indicating high uncertainty, a more cautious, phased release strategy might be warranted. Conversely, a sharp price surge forecast with high confidence would justify a larger, more immediate intervention. This transforms the reserve system from a blunt instrument into a precision tool for market calibration. Financial market regulation and the development of agricultural derivatives also stand to gain significantly. The high-fidelity forecasts can serve as a robust benchmark for valuing and structuring futures, options, and other price-risk management instruments. Regulators can use the model to monitor for dislocations between forecasted fundamental price paths and actual futures market prices, which may indicate the presence of excessive speculation or market manipulation. By identifying such anomalies early, regulatory bodies can enhance

market oversight and ensure that these financial markets fulfill their primary function of price discovery and risk transfer, rather than becoming sources of volatility themselves. This deepens market liquidity and provides agribusinesses with more reliable tools for hedging, thereby encouraging greater investment in the agricultural sector. From a macroeconomic and environmental sustainability perspective, the implications are far-reaching. The People's Bank of China and other monetary authorities rely on accurate inflation forecasts. Since food prices are a significant component of the Consumer Price Index (CPI), the superior forecasting of the Vegetable Basket index directly enhances the precision of broader inflation projections. This allows for more nuanced monetary policy, potentially avoiding overly restrictive measures that could stifle economic growth when food price inflation is transient. Environmentally, the ability to forecast favorable price windows can be leveraged to promote sustainable practices. For instance, if the model predicts strong prices, the government could couple this information with targeted subsidies or technical support for farmers to invest in water-saving irrigation systems or precision agriculture technologies during these periods of higher financial capacity. This aligns economic incentives with environmental goals, fostering a transition towards a more sustainable agricultural model. Finally, the model's intrinsic adaptability, ensured through Bayesian optimization, means it is not a static tool. As policies themselves–such as new subsidy schemes, trade agreements, or environmental regulations–are implemented, they become new data points that the model can assimilate. This creates a feedback loop where policy actions inform the model, and the model's updated forecasts, in turn, provide evidence for refining those very policies. In conclusion, the GPR forecasting framework presented here should be viewed not as an end in itself, but as a critical decision-support system. By providing accurate, probabilistic, and forward-looking insights, it empowers policymakers at all levels to move from a reactive posture to a strategic, evidence-based approach in guiding China's vital agricultural sector through the complexities of the 21st century. It transforms price prediction from an abstract statistical exercise into a concrete instrument for enhancing economic efficiency, social welfare, and environmental resilience.

9 Conclusion

This research confronts the complex forecasting challenge of China's daily wholesale agricultural commodity composite price indicators of the Vegetable Basket Project, spanning September 27, 2005, through April 10, 2025, utilizing Gaussian Process Regression (*GPR*). The methodological framework integrates a decile-stratified k-fold cross-validation architecture coupled with Bayesian hyperparameter calibration governed by the Expected Improvement Per Second Plus (*EIPSP*) acquisition function. Segmenting the dataset into subsets representing the complete spectrum of price realizations and conducting surrogate-guided exploration across an extensive hyperparameter domain establishes an optimal equilibrium between exhaustive calibration and computational tractability, thereby enhancing the extrapolative validity of the *GPR* system. To systematically encapsulate intrinsic nonlinear dynamics, model specification involves rigorous assessment of ten distinct covariance kernel formulations alongside four basis-function configurations. Throughout the in-sample calibration phase concluding February 16, 2022, each partition experiences autonomous refinement of critical parameters–specifically kernel length-scales, signal variance, noise

variance, and basis-function coefficients–employing the *EIPSP* acquisition methodology. Following identification of optimal configurations per segment, ten meticulously calibrated *GPR* surrogates–each demonstrating robust in-sample performance–are consolidated through arithmetic averaging. Predictive efficacy underwent rigorous evaluation across a 1,146-day out-of-sample horizon (February 17, 2022–April 10, 2025), characterized by regulatory transitions and economic volatility. The consolidated ensemble achieved a Relative Root-Mean-Square Error (*RRMSE*) of 0.2040%, Root-Mean-Square Error (*RMSE*) of 0.4841, Mean Absolute Error (*MAE*) of 0.3430, and Correlation Coefficient (*CC*) of 99.929%, accurately replicating both directional price movements and magnitude fluctuations within index trajectories. Beyond empirical validation, the *GPR* ensemble generates operationally actionable intelligence for diverse stakeholders: risk management professionals may utilize central projections and probabilistic intervals to optimize hedging strategies and mitigate volatility effects; institutional investors could integrate predictive distributions into long-horizon portfolio allocation and scenario analysis frameworks; regulatory bodies might leverage the model's nonparametric adaptability to simulate policy intervention outcomes. Core methodological advantages encompass structural flexibility–inherent in *GPR*'s capacity to assimilate regime transitions without structural modification–parametric versatility, enabled by heterogeneous kernel and mean-function selections permitting rapid integration of emergent information, and robustness assurances through decile-oriented validation combined with ensemble aggregation. This synthesis reduces overfitting susceptibility, diminishes anomaly sensitivity, and maintains temporal consistency. By unifying probabilistic surrogate modeling, systematic hyperparameter refinement, and ensemble forecasting, this investigation furnishes a transparent, resilient, and operationally practicable methodology for navigating the intrinsically intricate behavior of composite agricultural commodity price indicators. The framework's capacity to deliver quantified predictive uncertainty alongside computational efficiency renders it exceptionally valuable for central banking institutions and regulatory agencies operating within volatile commodity markets and evolving policy environments.

This study establishes the forecasting capability of a Bayesian-optimized GPR ensemble for the Vegetable Basket Project's composite price index. However, a recognized limitation of the current methodological design is its reliance on the autoregressive structure of the index itself, utilizing ten temporal lags as predictors while omitting potential control variables such as meteorological data, policy shift indicators, or input costs. This parsimonious approach was initially adopted to prioritize the capture of complex, intrinsic temporal dependencies and nonlinear dynamics through the GPR's kernel structure, thereby providing a robust baseline model. Nevertheless, we acknowledge that this design inherently attributes predictive power to the index's own history, potentially overlooking the fundamental drivers that cause the price movements it so accurately forecasts. Consequently, the model, in its present form, may be less adept at forecasting the impact of unprecedented exogenous shocks that fall outside the historical pattern of the series itself. This presents a compelling avenue for future research. A logical and critical next step is to augment the GPR framework by integrating a curated set of exogenous variables. Future work will focus on incorporating datasets representing climatic conditions (e.g., drought indices, temperature anomalies), discrete policy events (e.g., implementation of new subsidies or trade agreements represented as dummy variables), and macroeconomic factors (e.g., fertilizer price indices, transportation costs). The inherent flexibility of the GPR covariance function is

particularly well-suited for this task, as additive kernels could be constructed to seamlessly blend the influence of these diverse data types. Such an enriched model might not only aim to improve predictive accuracy further but, more importantly, might transition from a purely statistical forecasting tool to a more causal, explanatory framework. This would allow for a quantitative assessment of how specific factors–like an extreme weather event or a strategic reserve release–propagate through the system to affect the composite price, thereby offering even deeper, more actionable insights for policymakers seeking to understand and manage the underlying drivers of price volatility in China's agricultural sector.

While this study has focused primarily on the point forecast accuracy of the Gaussian Process Regression (GPR) ensemble–demonstrating its superior performance in tracking the central tendency of the Vegetable Basket Project index–we acknowledge an underutilization of a key GPR strength: its innate, analytically tractable uncertainty quantification. Indeed, a principal advantage of the Bayesian nonparametric paradigm is its capacity to generate full predictive distributions, providing not just a point estimate but also well-calibrated credible intervals that quantify forecast uncertainty. Our current analysis, centered on metrics like RMSE and MAE, does not explicitly validate the reliability of these predictive intervals. A critical and valuable direction for future research will be to rigorously evaluate the calibration of these intervals using metrics such as the prediction interval coverage probability (PICP), which assesses the empirical frequency that the true observation falls within a given credible interval, and the mean scoring interval length, which measures their sharpness. For instance, a 95% posterior predictive interval should contain the true value approximately 95% of the time in out-of-sample testing; significant deviations would indicate miscalibration. Undertaking this analysis would transform the model from a high-accuracy forecasting tool into a robust risk-assessment system. Such probabilistically calibrated forecasts are indispensable for advanced policy planning, enabling stakeholders to move beyond "most likely" scenarios and formally evaluate tail risks. Policymakers could, for example, use these intervals to estimate the probability of the price index breaching a critical threshold that would trigger strategic reserve releases, or agribusinesses could more accurately quantify Value-at-Risk. Therefore, future work could include a comprehensive analysis of interval forecasting performance, ensuring that the model's uncertainty estimates are as reliable as its point forecasts, thereby fully leveraging the Bayesian machinery of GPR for decision-making under uncertainty.

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Declarations

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